Statistics

Chapter 13 — Instructor Solutions Manual

October 21, 2026 • Total Exercises: 64

PROBLEM STATEMENT

Chapter 13, Page 470

Step 1

Given:

$$n = \text{Sample size} = 5$$

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{4+5+3+6+10}{5} = 5.6$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{4+6+5+7+7}{5} = 5.8$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(4-5.6)^2 + ... + (10-5.6)^2}{5-1}} \approx 2.7019$$
$$s_y = \sqrt{\frac{(4-5.8)^2 + ... + (7-5.8)^2}{5-1}} \approx 1.3038$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X-\overline{X}$	$Y-\overline{Y}$	$(X-\overline{X})(Y-\overline{Y})$
4	4	-1.6	-1.8	2.88
5	6	-0.6	0.2	-0.12
3	5	-2.6	-0.8	2.08
6	7	0.4	1.2	0.48
10	7	4.4	1.2	5.28
			Total	10.6

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{10.6}{(5-1)(2.7019)(1.3038)} \\ &\approx 0.7522 \end{aligned}$$

Step 3

If r is positive, then there is a positive linear correlation. If r is negative, then there is a negative linear correlation.

If $0 < |r| < 0.5$, then the correlation is weak. If $0.5 < |r| < 0.8$, then the correlation is moderate. If $0.8 < |r| < 1$, then the correlation is strong.

Since r is between 0.5 and 0.8 and positive, there is a moderately strong positive relationship between X and Y.

Result

$$r = 0.7522$$

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Solution A

Step 1

In this problem, we need to calculate the correlation coefficient r between the given x and y data sets. We also need to interpret what the calculated correlation coefficient means.

Step 2

To solve this problem we will use the **Pearson correlation coefficient for a sample** equation given in the book:

$$r = \frac{\sum(x - \overline{x})(y - \overline{y})}{(n-1)s_x s_y}$$

where:

- x and y are the given samples,
- $\overline{x}$ and $\overline{y}$ are their mean values,
- n is the sample size and
- s_x and s_y are standard deviations of the samples.

Correlation coefficient is a number that measures a linear correlation between the two given samples. The value of the coefficient can be between -1 and 1. If the value is zero, or close to zero then the samples are **not correlated**. As the absolute value increases so does the correlation between the data sets. The closer the value is to 1 the correlation is stronger. **Positive values** mean a **positive correlation** and **negative values** mean a **negative or inverse correlation**.

Step 3

The given data sets are:

x	y
5	13
3	15
6	7
3	12
4	13
4	11
6	9
8	5

Step 4

Using the appropriate software we can determine the required **mean** and **standard deviation** values:

$$\overline{x} = 4.875$$

$$\overline{y} = 10.625$$

$$s_x = 1.615$$

$$s_y = 3.160$$

The number of values n can easily be determined by counting the given values:

$$n = 8$$

Step 5

Next, we can again use the appropriate software to calculate the sum in the numerator of the equation:

$$\sum(x-\overline{x})(y-\overline{y}) = -36.375$$

Finally, we can substitute all the values back in the equation and determine the **correlation coefficient**:

$$\begin{aligned} r &= \frac{\sum(x-\overline{x})(y-\overline{y})}{(n-1)s_x s_y} \\ &= \frac{-36.375}{(8-1)\cdot 1.615 \cdot 3.160} \\ &= \boxed{-0.891} \end{aligned}$$

Step 6

The correlation coefficient between the given data sets x and y is -0.891. The coefficient is negative meaning that the given data sets are **inversely correlated**. The value of the coefficient is close to -1 meaning that the correlation is **strong**.

Result

$$r = -0.891$$

Step 1

Given:

X	5	3	6	3	4	4	6	8
Y	13	15	7	12	13	11	9	5

Step 1: Find $X \cdot Y$, X^2 and Y^2 as it was done in the table below.

X	Y	X· Y	X·X	Y·Y
5	13	65	25	169
3	15	45	9	225
6	7	42	36	49
3	12	36	9	144
4	13	52	16	169
4	11	44	16	121
6	9	54	36	81
8	5	40	64	25

Step 2: Find the sum of every column to get:

$$\sum X = 39$$

$$\sum Y = 85$$

$$\sum X \cdot Y$$

$$\sum X^2 = 211$$

$$\sum Y^2 = 983$$

Step 3: Use the following formula to work out the correlation coefficient.

$$r = \frac{n \cdot \sum XY - \sum X \cdot \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2] \cdot [n \sum Y^2 - (\sum Y)^2]}}$$

$$r = \frac{8 \cdot 378 - 39 \cdot 85}{\sqrt{[8 \cdot 211 - 39^2] \cdot [8 \cdot 983 - 85^2]}} \approx -0.8908$$

🎯 Result

$$r = -0.8908$$

Solution B

Step 1

In this problem, we need to calculate the correlation coefficient r between the given x and y data sets. We also need to interpret what the calculated correlation coefficient means.

Step 2

To solve this problem we will use the **Pearson correlation coefficient for a sample** equation given in the book:

$$r = \frac{\sum(x - \overline{x})(y - \overline{y})}{(n-1)s_x s_y}$$

where:

- x and y are the given samples,
- $\overline{x}$ and $\overline{y}$ are their mean values,
- n is the sample size and
- s_x and s_y are standard deviations of the samples.

Correlation coefficient is a number that measures a linear correlation between the two given samples. The value of the coefficient can be between -1 and 1. If the value is zero, or close to zero then the samples are **not correlated**. As the absolute value increases so does the correlation between the data sets. The closer the value is to 1 the correlation is stronger. **Positive values** mean a **positive correlation** and **negative values** mean a **negative or inverse correlation**.

Step 3

The given data sets are:

x	y
5	13
3	15
6	7
3	12
4	13
4	11
6	9
8	5

Step 4

Using the appropriate software we can determine the required **mean** and **standard deviation** values:

$$\overline{x} = 4.875$$

$$\overline{y} = 10.625$$

$$s_x = 1.615$$

$$s_y = 3.160$$

The number of values n can easily be determined by counting the given values:

$$n = 8$$

Step 5

Next, we can again use the appropriate software to calculate the sum in the numerator of the equation:

$$\sum (x - \overline{x})(y - \overline{y}) = -36.375$$

Finally, we can substitute all the values back in the equation and determine the **correlation coefficient**:

$$\begin{aligned} r &= \frac{\sum (x - \overline{x})(y - \overline{y})}{(n-1)s_x s_y} \\ &= \frac{-36.375}{(8-1) \cdot 1.615 \cdot 3.160} \\ &= \boxed{-0.891} \end{aligned}$$

Step 6

The correlation coefficient between the given data sets x and y is -0.891. The coefficient is negative meaning that the given data sets are **inversely correlated**. The value of the coefficient is close to -1 meaning that the correlation is **strong**.

🎯 Result

$$r = -0.891$$

Step 1

Given:

X	5	3	6	3	4	4	6	8
Y	13	15	7	12	13	11	9	5

Step 1: Find $X \cdot Y$, X^2 and Y^2 as it was done in the table below.

X	Y	X· Y	X·X	Y·Y
5	13	65	25	169
3	15	45	9	225
6	7	42	36	49
3	12	36	9	144
4	13	52	16	169
4	11	44	16	121
6	9	54	36	81
8	5	40	64	25

Step 2: Find the sum of every column to get:

$$\sum X = 39$$

$$\sum Y = 85$$

$$\sum X \cdot Y$$

$$\sum X^2 = 211$$

$$\sum Y^2 = 983$$

Step 3: Use the following formula to work out the correlation coefficient.

$$r = \frac{n \cdot \sum XY - \sum X \cdot \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2] \cdot [n \sum Y^2 - (\sum Y)^2]}}$$

$$r = \frac{8 \cdot 378 - 39 \cdot 85}{\sqrt{[8 \cdot 211 - 39^2] \cdot [8 \cdot 983 - 85^2]}} \approx -0.8908$$

Result

$$r = -0.8908$$

PROBLEM STATEMENT

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Step 1

Given:

$$n = \text{Sample size} = 5$$

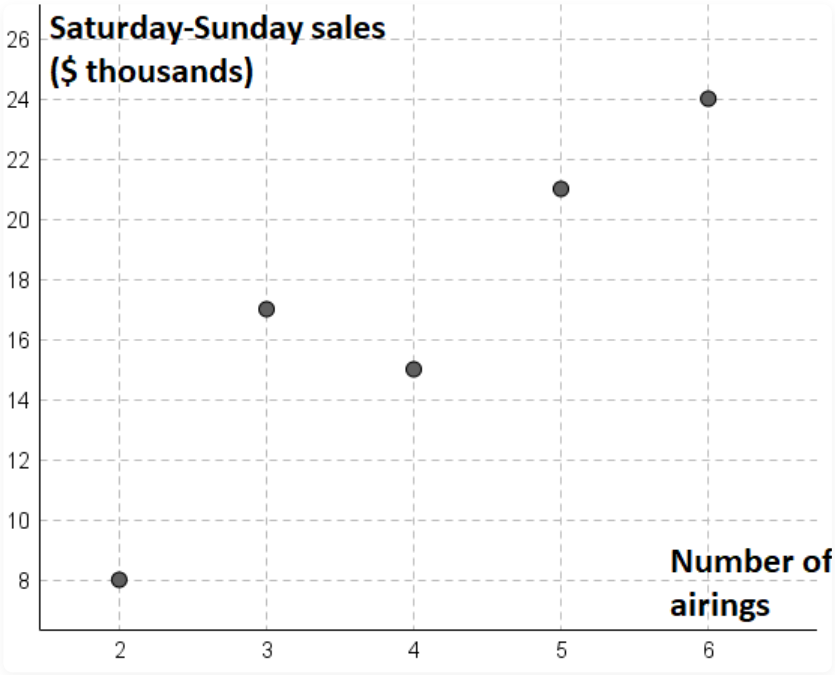
(a) A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

We expect the sales to increases as the number of airings increase, which implies that the independent variable is the number of airings and the dependent variable is the sales.

Step 2

(b) **Scatter diagram**

Number of airings is on the horizontal axis and Saturday-Sunday Sales is on the vertical axis.



Step 3

(c) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{4+2+5+6+3}{5} = 4$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{15+8+21+24+17}{5} = 17$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(4-4)^2 + ... + (3-4)^2}{5-1}} \approx 1.5811$$
$$s_y = \sqrt{\frac{(15-17)^2 + ... + (17-17)^2}{5-1}} \approx 6.1237$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
4	15	0	-2	0
2	8	-2	-9	18
5	21	1	4	4
6	24	2	7	14
3	17	-1	0	0
			Total	36

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{36}{(5-1)(1.5811)(6.1237)} \\ &\approx 0.9295 \end{aligned}$$

Step 5

(d) If r is positive, then there is a positive linear correlation. If r is negative, then there is a negative linear correlation.

If $0 < |r| < 0.5$, then the correlation is weak. If $0.5 < |r| < 0.8$, then the correlation is moderate. If $0.8 < |r| < 1$, then the correlation is strong.

Since r is larger than 0.8 and positive, there is a strong positive relationship between number of airings and sales.

🎯 Result

(a) Saturday-Sunday Sales

(b) See scatter diagram

(c) $r = 0.9295$

(d) There is a strong positive relationship between number of airings and sales.

PROBLEM STATEMENT

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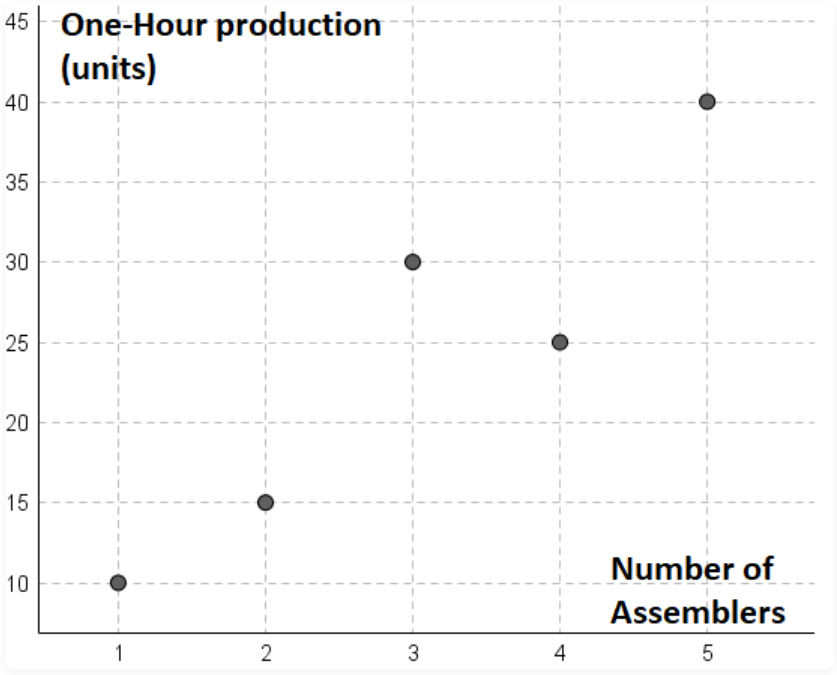
Step 1

Given:

$$n = \text{Sample size} = 5$$

(a) **Scatter diagram**

Number of assemblers is on the horizontal axis and One-hour production is on the vertical axis.



Step 2

(b) The pattern in the scatter diagram slopes upwards, which indicates that there is a (positive) relationship between the number of assemblers and production.

Step 3

(c) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2+4+1+5+3}{5} = 3$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{15+25+10+40+30}{5} = 24$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(2-3)^2 + ... + (3-3)^2}{5-1}} \approx 1.5811$$
$$s_y = \sqrt{\frac{(15-24)^2 + ... + (30-24)^2}{5-1}} \approx 11.9373$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
2	15	-1	-9	9
4	25	1	1	1
1	10	-2	-14	28
5	40	2	16	32
3	30	0	6	0
			Total	70

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{70}{(5-1)(1.5811)(11.9373)} \\ &\approx 0.9272 \end{aligned}$$

Result

(a) See scatter diagram

(b) Yes

(c) $r = 0.9272$

PROBLEM STATEMENT

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Step 1

Given:

$$n = \text{Sample size} = 8$$

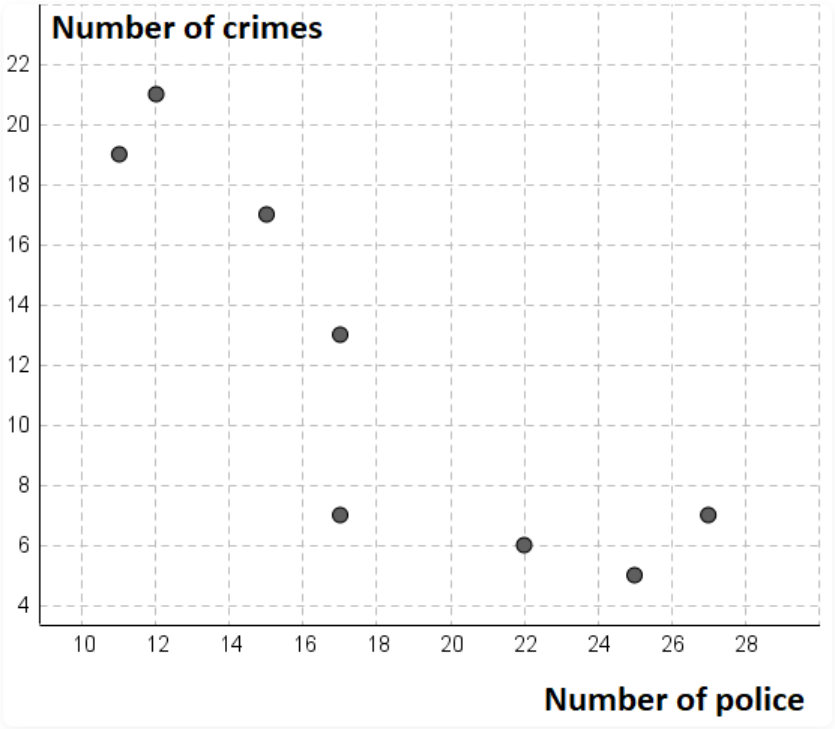
(a) A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

We expect the number of crimes to decrease as the number of police increase, which implies that the independent variable is the number of police and the dependent variable is the number of crimes.

Step 2

(b) **Scatter diagram**

Number of crimes is on the horizontal axis and Number of police is on the vertical axis.



Step 3

(c) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{15 + 17 + 25 + 17 + 17 + 12 + 11 + 22}{8} = 18.25$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{17 + 13 + 5 + 7 + 7 + 21 + 19 + 6}{8} = 11.875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(15 - 18.25)^2 + ... + (22 - 18.25)^2}{8 - 1}} \approx 5.8737$$
$$s_y = \sqrt{\frac{(17 - 11.875)^2 + ... + (6 - 11.875)^2}{8 - 1}} \approx 6.4462$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
15	17	-3.25	5.125	-16.65625
17	13	-1.25	1.125	-1.40625
25	5	6.75	-6.875	-46.40625
27	7	8.75	-4.875	-42.65625
17	7	-1.25	-4.875	6.09375
12	21	-6.25	9.125	-57.03125
11	19	-7.25	7.125	-51.65625
22	6	3.75	-5.875	-22.03125
			Total	-231.75

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-231.75}{(8-1)(5.8737)(6.4462)} \\ &\approx -0.8744 \end{aligned}$$

Step 5

(d) If r is positive, then there is a positive linear correlation. If r is negative, then there is a negative linear correlation.

If $0 < |r| < 0.5$, then the correlation is weak. If $0.5 < |r| < 0.8$, then the correlation is moderate. If $0.8 < |r| < 1$, then the correlation is strong.

Since r is smaller than -0.8 and negative, there is a strong negative relationship between the number of police and the number of crimes.

It is not surprising that the correlation coefficient is negative, because we expect the number of crimes to decrease as the number of police increase.

Result

(a) Number of crimes

(b) See scatter diagram

(c) $r = -0.8744$

(d) There is a strong negative relationship between the number of police and the number of crimes.

No

PROBLEM STATEMENT

Chapter 13, Page 472

Step 1

Given:

$$n = \text{Sample size} = 12$$

(a) **Scatter diagram**

Age is on the horizontal axis and Selling price is on the vertical axis.



Step 2

(b) Formula correlation coefficient:

$$r = \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{9 + 7 + 11 + ... + 6}{12} = 8.9167$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{8.1 + 6 + 3.6 + ... + 6}{12} = 7.475$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(9 - 8.9167)^2 + ... + (6 - 8.9167)^2}{12 - 1}} \approx 2.2344$$
$$s_y = \sqrt{\frac{(8.1 - 7.475)^2 + ... + (6 - 7.475)^2}{12 - 1}} \approx 2.7971$$

Step 3

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
9	8.1	0.0833	0.625	0.0521
7	6	-1.9167	-1.475	2.8271
11	3.6	2.0833	-3.875	-8.0728
12	4	3.0833	-3.475	-10.7145
8	5	-0.9167	-2.475	2.2688
7	10	-1.9167	2.525	-4.8397
8	8	-0.9167	0.525	-0.4813
11	11	2.0833	3.525	7.3436
10	10	1.0833	2.525	2.7353
12	12	3.0833	4.525	13.9519
6	6	-2.9167	-1.475	4.3021
6	6	-2.9167	-1.475	4.3021
			Total	13.675

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{13.675}{(12-1)(2.2344)(6.1237)} \\ &\approx 0.1989 \end{aligned}$$

Step 4

(d) If r is positive, then there is a positive linear correlation. If r is negative, then there is a negative linear correlation.

If $0 < |r| < 0.5$, then the correlation is weak. If $0.5 < |r| < 0.8$, then the correlation is moderate. If $0.8 < |r| < 1$, then the correlation is strong.

Since r is smaller than 0.5 and positive, there is a weak positive relationship between age and selling price.

The question about whether it is surprising that the correlation coefficient is negative appears to be incorrect, as the correlation coefficient is actually positive.

🎯 Result

(a) See scatter diagram

(b) $r = 0.1989$

(c) There is a weak positive relationship between age and selling price.

PROBLEM STATEMENT

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Step 1

Given:

$$n = \text{Sample size} = 12$$
$$r = \text{Correlation coefficient} = 0.32$$
$$\alpha = \text{Significance level} = 0.05$$
$$H_0 : \rho \leq 0$$
$$H_1 : \rho > 0$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 12 - 2 = 10$ and in the column with $\alpha = 0.05$.

$$t = 1.812$$

The rejection region then contains all values greater than 1.812 and thus we reject H_0 when the t-value is larger than 1.812.

Step 2

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.32\sqrt{12-2}}{\sqrt{1-0.32^2}} \approx 1.068$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 1.068 < 1.812 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the correlation in the population is greater than 0.

🎯 Result

No

PROBLEM STATEMENT

Chapter 13, Page 475

Step 1

Given:

$$\begin{aligned} n &= \text{Sample size} = 15 \\ r &= \text{Correlation coefficient} = -0.46 \\ \alpha &= \text{Significance level} = 0.05 \\ H_0 &: \rho \geq 0 \\ H_1 &: \rho < 0 \end{aligned}$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 15 - 2 = 14$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.771$$

The rejection region then contains all values smaller than -1.771 and thus we reject H_0 when the t-value is smaller than -1.771.

Step 2

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.46\sqrt{15-2}}{\sqrt{1-(-0.46)^2}} \approx -1.868$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -1.868 < -1.771 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the correlation in the population is less than zero.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 475

Step 1

Given:

$$n = \text{Sample size} = 20$$
$$r = \text{Correlation coefficient} = 0.78$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: Correlation is greater than zero

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 20 - 2 = 18$ and in the column with $\alpha = 0.01$.

$$t = 2.552$$

The rejection region then contains all values greater than 2.552 and thus we reject H_0 when the t-value is larger than 2.552.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.78\sqrt{20-2}}{\sqrt{1-0.78^2}} \approx 5.288$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 5.288 > 2.552 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the correlation in the population is greater than zero.

Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 475

Step 1

Given:

$$n = \text{Sample size} = 20$$
$$r = \text{Correlation coefficient} = 0.86$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Positive correlation

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 20 - 2 = 18$ and in the column with $\alpha = 0.05$.

$$t = 1.734$$

The rejection region then contains all values greater than 1.734 and thus we reject H_0 when the t-value is larger than 1.734.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.86\sqrt{20-2}}{\sqrt{1-0.86^2}} \approx 7.150$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 7.150 > 1.734 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a positive correlation in the population.

Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 475

Step 1

Given:

$$n = \text{Sample size} = 15$$
$$r = \text{Correlation coefficient} = 0.667$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 15 - 2 = 13$ and in the column with $\alpha = 0.01$.

$$t = 2.650$$

The rejection region then contains all values greater than 2.650 and thus we reject H_0 when the t-value is larger than 2.650.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.667\sqrt{15-2}}{\sqrt{1-0.667^2}} \approx 3.228$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 3.228 > 2.650 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a positive association in the population between the two variables.

Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 475

Step 1

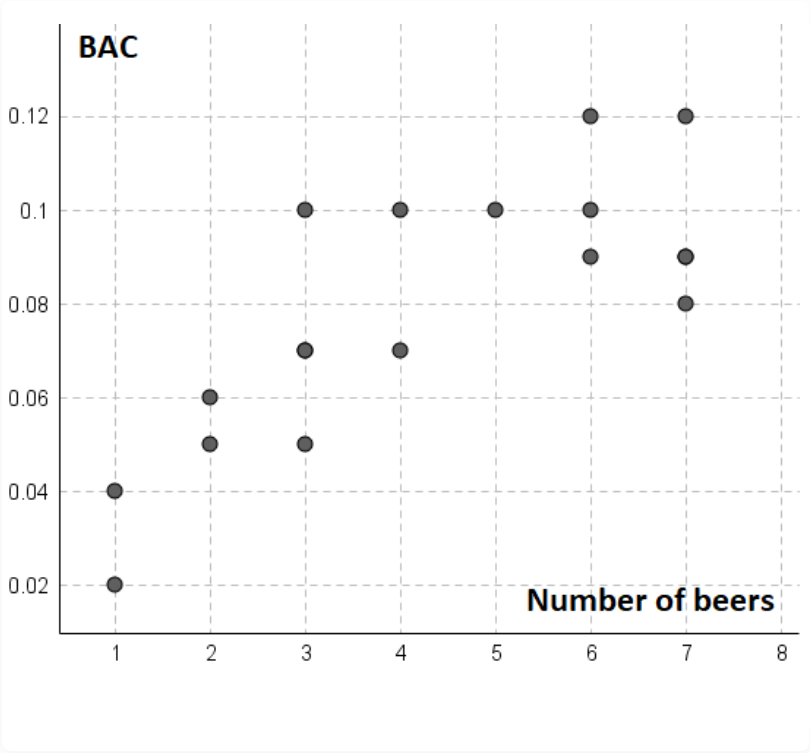
Given:

$$n = \text{Sample size} = 18$$
$$\alpha = \text{Significance level} = 0.01$$

(a)

Scatter diagram

Number of beers is on the horizontal axis and BAC is on the vertical axis.



Step 2

The relationship appears to be strong, because the points in the scatterplot are not spread very far apart.

The pattern in the scatter diagram slopes upwards, which implies that there is a positive relationship between the number of beers consumed and the BAC.

Step 3

(b) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{6 + 7 + 7 + ... + 2 + 1}{18} = 4.2778$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{0.1 + 0.09 + 0.09 + ... + 0.05 + 0.02}{18} = 0.0789$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(6 - 4.2778)^2 + ... + (1 - 4.2778)^2}{18 - 1}} \approx 2.1367$$
$$s_y = \sqrt{\frac{(0.1 - 0.0789)^2 + ... + (0.02 - 0.0789)^2}{18 - 1}} \approx 0.0274$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
6	0.1	1.7222	0.0211	0.0363
7	0.09	2.7222	0.0111	0.0302
7	0.09	2.7222	0.0111	0.0302
4	0.1	-0.2778	0.0211	-0.0059
5	0.1	0.7222	0.0211	0.0152
3	0.07	-1.2778	-0.0089	0.0114
3	0.1	-1.2778	0.0211	-0.0270
6	0.12	1.7222	0.0411	0.0708
6	0.09	1.7222	0.0111	0.0191
3	0.07	-1.2778	-0.0089	0.0114
3	0.05	-1.2778	-0.0289	0.0369
7	0.08	2.7222	0.0011	0.0030
1	0.04	-3.2778	-0.0389	0.1275
4	0.07	-0.2778	-0.0089	0.0025
2	0.06	-2.2778	-0.0189	0.0431
7	0.12	2.7222	0.0411	0.1119
2	0.05	-2.2778	-0.0289	0.0658
1	0.02	-3.2778	-0.0589	0.1931
			Total	0.7756

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{0.7756}{(18-1)(2.1367)(0.0274)} \\ &\approx 0.7788 \end{aligned}$$

Step 5

(c) Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$\begin{aligned} &H_0 : \rho \leq 0 \text{ (There is no linear correlation)} \\ &H_a : \rho > 0 \text{ (There is a positive linear correlation)} \end{aligned}$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 18 - 2 = 16$ and in the column with $\alpha = 0.01$.

$$t = 2.583$$

The rejection region then contains all values greater than 2.583 and thus we reject H_0 when the t-value is larger than 2.583.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.7788\sqrt{18-2}}{\sqrt{1-0.7788^2}} \approx 4.966$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 4.966 > 2.583 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a positive relationship in the population between the number of beers consumed and the BAC.

Step 7

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 4.966 in the row $df = n - 2 = 18 - 2 = 16$ (one-tailed):

$$P < 0.0005$$

Note: When using technology, then you obtain the P-value 0.00007.

🎯 **Result**

(a) Strong, positive

(b) $r = 0.7788$

(c) Yes, $P < 0.0005$

PROBLEM STATEMENT

Chapter 13, Page 481

Step 1

Given:

$$n = \text{Sample size} = 5$$

(a)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{4+5+3+6+10}{5} = 5.6$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{4+6+5+7+7}{5} = 5.8$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(4-5.6)^2 + ... + (10-5.6)^2}{5-1}} \approx 2.7019$$
$$s_y = \sqrt{\frac{(4-5.8)^2 + ... + (7-5.8)^2}{5-1}} \approx 1.3038$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
4	4	-1.6	-1.8	2.88
5	6	-0.6	0.2	-0.12
3	5	-2.6	-0.8	2.08
6	7	0.4	1.2	0.48
10	7	4.4	1.2	5.28
			Total	10.6

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1) s_x s_y} \\ &= \frac{10.6}{(5-1)(2.7019)(1.3038)} \\ &\approx 0.7522 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r \frac{s_y}{s_x} = 0.7522 \frac{1.3038}{2.7019} = 0.3630$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 5.8 - 0.3630 \cdot 5.6 = 3.7671$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 3.7671 + 0.3630x$$

Step 4

(b) Evaluate the regression equation found in part (a) at $x = 7$:

$$\hat{y} = a + bx = 3.7671 + 0.3630(7) = 6.3082$$

Result

(a) $\hat{y} = 3.7671 + 0.3630x$

(b) $\hat{y} = 6.3082$

PROBLEM STATEMENT

Chapter 13, Page 481

Step 1

Given:

$$n = \text{Sample size} = 8$$

(a)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{5 + 3 + 6 + ... + 8}{8} = 4.875$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{13 + 15 + 7 + ... + 5}{8} = 10.625$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(5 - 4.875)^2 + ... + (8 - 4.875)^2}{8 - 1}} \approx 1.7269$$
$$s_y = \sqrt{\frac{(13 - 10.625)^2 + ... + (5 - 10.625)^2}{8 - 1}} \approx 3.3780$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
5	13	0.125	2.375	0.296875
3	15	-1.875	4.375	-8.203125
6	7	1.125	-3.625	-4.078125
3	12	-1.875	1.375	-2.57812
4	13	-0.875	2.375	-2.078125
4	11	-0.875	0.375	-0.328125
6	9	1.125	-1.625	-1.828125
8	5	3.125	-5.625	-17.5781
			Total	**-36.375**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-36.375}{(8-1)(1.7269)(3.3780)} \\ &\approx -0.8908 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.8908\frac{3.3780}{1.7269} = -1.7425$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 10.625 - (-1.7425) \cdot 4.875 = 19.1198$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 19.1198 - 1.7425x$$

Step 4

(b) Evaluate the regression equation found in part (a) at $x = 7$:

$$\hat{y} = a + bx = 19.1198 - 1.7425(7) = 6.9222$$

🎯 Result

(a) $\hat{y} = 19.1198 - 1.7425x$

(b) $\hat{y} = 6.9222$

PROBLEM STATEMENT

Chapter 13, Page 482

Step 1

Given:

$$n = \text{Sample size} = 10$$

(a)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{12 + 9 + 14 + ... + 7}{10} = 9.1$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{9 + 7 + 10 + ... + 7}{10} = 7.4$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(12 - 9.1)^2 + ... + (7 - 9.1)^2}{10 - 1}} \approx 2.7264$$
$$s_y = \sqrt{\frac{(9 - 7.4)^2 + ... + (7 - 7.4)^2}{10 - 1}} \approx 2.0111$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
12	9	2.9	1.6	4.64
9	7	-0.1	-0.4	0.04
14	10	4.9	2.6	12.74
6	5	-3.1	-2.4	7.44
10	8	0.9	0.6	0.54
8	6	-1.1	-1.4	1.54
10	8	0.9	0.6	0.54
10	10	0.9	2.6	2.34
5	4	-4.1	-3.4	13.94
7	7	-2.1	-0.4	0.84
			Total	**44.6**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{44.6}{(10-1)(2.7264)(2.0111)} \\ &\approx 0.9038 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9038\frac{2.0111}{2.7264} = 0.6667$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 7.4 - 0.6667 \cdot 9.1 = 1.3333$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 1.3333 + 0.6667x$$

Step 4

(b) Evaluate the regression equation found in part (a) at $x = 6$:

$$\hat{y} = a + bx = 1.3333 + 0.6667(6) = 5.3333$$

Thus we expect a six-room house to use 5.3333 thousand kilowatt-hours.

🎯 Result

(a) $\hat{y} = 1.3333 + 0.6667x$

(b) $\hat{y} = 5.3333$ thousand kilowatt-hours

PROBLEM STATEMENT

Chapter 13, Page 482

Step 1

Given:

$$n = \text{Sample size} = 10$$

(a)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{14 + 12 + 20 + ... + 50}{10} = 33.4$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{24 + 14 + 28 + ... + 110}{10} = 61.1$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(14 - 33.4)^2 + ... + (50 - 33.4)^2}{10 - 1}} \approx 17.6836$$
$$s_y = \sqrt{\frac{(24 - 61.1)^2 + ... + (110 - 61.1)^2}{10 - 1}} \approx 39.7896$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
14	24	-19.4	-37.1	719.74
12	14	-21.4	-47.1	1007.94
20	28	-13.4	-33.1	443.54
16	30	-17.4	-31.1	541.14
46	80	12.6	18.9	238.14
23	30	-10.4	-31.1	323.44
48	90	14.6	28.9	421.94
50	85	16.6	23.9	396.74
55	120	21.6	58.9	1272.24
50	110	16.6	48.9	812.74
			Total	**6176.6**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{6176.6}{(10-1)(17.6836)(39.7896)} \\ &\approx 0.9754 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9754\frac{39.7896}{17.6836} = 2.1946$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 61.1 - 2.1946 \cdot 33.4 = -12.2010$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -12.2010 + 2.1946x$$

Step 4

(b) Evaluate the regression equation found in part (a) at $x = 40$:

$$\hat{y} = a + bx = -12.2010 + 2.1946(40) = 75.5846$$

Thus the estimated sales are 75.5846 thousand dollars.

🎯 Result

(a) $\hat{y} = -12.2010 + 2.1946x$

(b) $\hat{y} = 75.5846$ thousand dollars

PROBLEM STATEMENT

Chapter 13, Page 482

Step 1

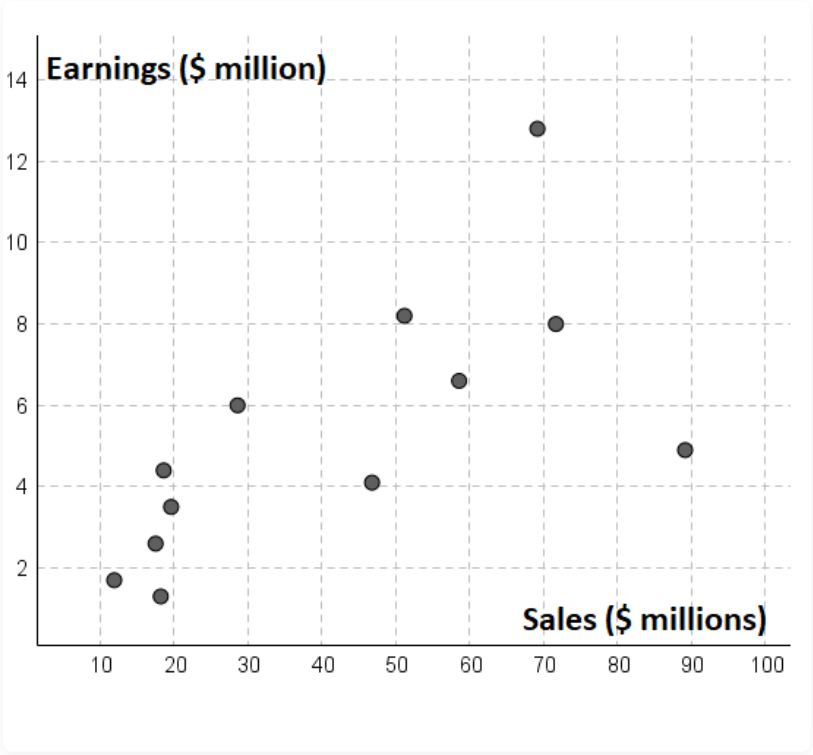
Given:

$$n = \text{Sample size} = 5$$

(a)

Scatter diagram

Sales is on the horizontal axis and Earnings is on the vertical axis.



Step 2

(b)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{89.2 + 18.6 + ... + 69.2}{12} = 41.7583$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{4.9 + 4.4 + ... + 12.8}{12} = 5.3417$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(89.2 - 41.7583)^2 + ... + (69.2 - 41.7583)^2}{12 - 1}} \approx 26.1706$$
$$s_y = \sqrt{\frac{(4.9 - 5.3417)^2 + ... + (12.8 - 5.3417)^2}{12 - 1}} \approx 3.2475$$

Step 3

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
89.2	4.9	47.4417	-0.4417	-20.9550
18.6	4.4	-23.1583	-0.9417	21.8082
18.2	1.3	-23.5583	-4.0417	95.2156
71.7	8	29.9417	2.6583	79.5940
58.6	6.6	16.8417	1.2583	21.1919
46.8	4.1	5.0417	-1.2417	-6.2603
17.5	2.6	-24.2583	-2.7417	66.5090
11.9	1.7	-29.8583	-3.6417	108.7350
19.6	3.5	-22.1583	-1.8417	40.8089
51.2	8.2	9.4417	2.8583	26.9872
28.6	6	-13.1583	0.6583	-8.6621
69.2	12.8	27.4417	7.4583	204.6684
			Total	**629.6408**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{629.6408}{(12-1)(26.1706)(3.2475)} \\ &\approx 0.6735 \end{aligned}$$

Step 4

(c)

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.6735\frac{3.2475}{26.1706} = 0.0836$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 5.3417 - 0.0836 \cdot 41.7583 = 1.8517$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 1.8517 + 0.0836x$$

Step 5

(d) Evaluate the regression equation found in part (c) at $x = 50.0$:

$$\hat{y} = a + bx = 1.8517 + 0.0836(50.0) = 6.0305$$

Thus the estimated earnings is 6.0305 million dollars.

Result

(a) Scatter diagram

(b) $r = 0.6735$

(c) $\hat{y} = 1.8517 + 0.0836x$

(b) $\hat{y} = 6.0305$ million dollars

PROBLEM STATEMENT

Chapter 13, Page 482

Step 1

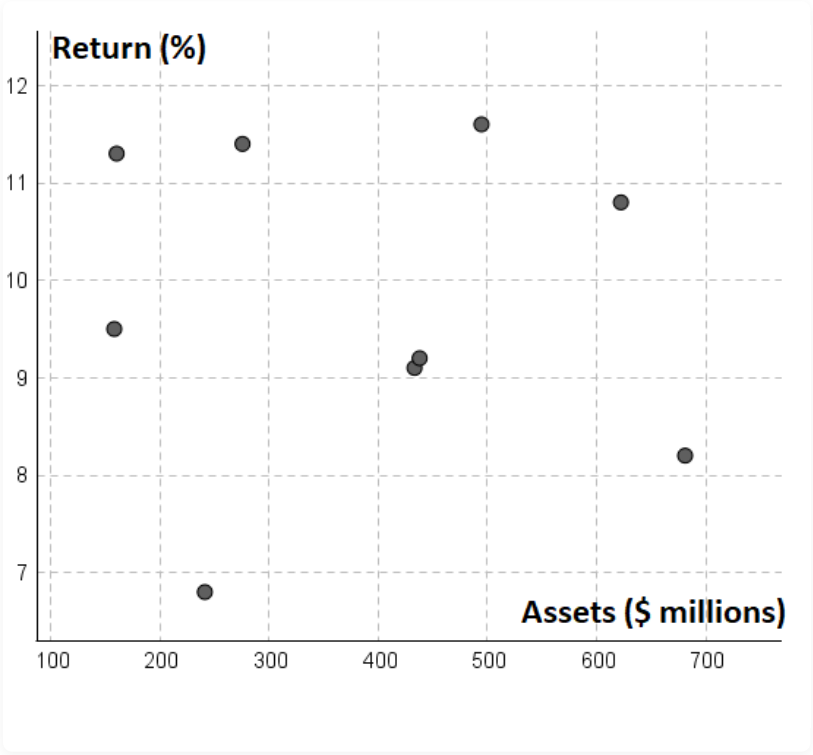
Given:

$$n = \text{Sample size} = 9$$

(a)

Scatter diagram

Assets is on the horizontal axis and return is on the vertical axis.



Step 2

(b)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{622.2 + 160.4 + ... + 241.3}{9} = 389.3889$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{10.8 + 11.3 + ... + 6.8}{9} = 9.7667$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(622.2 - 389.3889)^2 + ... + (241.3 - 389.3889)^2}{9 - 1}} \approx 192.1108$$
$$s_y = \sqrt{\frac{(10.8 - 9.7667)^2 + ... + (6.8 - 9.7667)^2}{9 - 1}} \approx 1.6409$$

Step 3

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
622.2	10.8	232.8111	1.0333	240.5637
160.4	11.3	-228.9889	1.5333	-351.1087
275.7	11.4	-113.6889	1.6333	-185.6881
433.2	9.1	43.8111	-0.6667	-29.2089
437.9	9.2	48.5111	-0.5667	-27.4912
494.5	11.6	105.1111	1.8333	192.7002
158.3	9.5	-231.0889	-0.2667	61.6314
681	8.2	291.6111	-1.5667	-456.8671
241.3	6.8	-148.0889	-2.9667	439.3353
			Total	**-116.1333**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-116.1333}{(9-1)(192.1108)(1.6409)} \\ &\approx -0.0461 \end{aligned}$$

Step 4

(c) If r is positive, then there is a positive linear correlation. If r is negative, then there is a negative linear correlation.

If $0 < |r| < 0.5$, then the correlation is weak. If $0.5 < |r| < 0.8$, then the correlation is moderate. If $0.8 < |r| < 1$, then the correlation is strong.

Since the correlation coefficient $r = -0.0461$ is very close to 0 and negative, there is a weak, negative, linear relationship between assets and return.

Step 5

(d)

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.0461\frac{1.6409}{192.1108} = -0.0004$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 9.7667 - (-0.0004) \cdot 389.3889 = 9.9198$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 9.9198 - 0.0004x$$

Step 6

(e) Evaluate the regression equation found in part (a) at $x = 400.0$:

$$\hat{y} = a + bx = 9.9198 - 0.0004(400.0) = 9.7625$$

The estimate five-year rate of return is 9.7625%.

Result

(a) Scatter diagram

(b) $r = -0.0461$

(c) There is a weak, negative, linear relationship between assets and return.

(d) $\hat{y} = 9.9198 - 0.0004x$

(e) $\hat{y} = 9.7625\%$

PROBLEM STATEMENT

Chapter 13, Page 483

Step 1

Given:

$$n = \text{Sample size} = 8$$

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{15 + 17 + 25 + 17 + 17 + 12 + 11 + 22}{8} = 18.25$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{17 + 13 + 5 + 7 + 7 + 21 + 19 + 6}{8} = 11.875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(15 - 18.25)^2 + ... + (22 - 18.25)^2}{8 - 1}} \approx 5.8737$$
$$s_y = \sqrt{\frac{(17 - 11.875)^2 + ... + (6 - 11.875)^2}{8 - 1}} \approx 6.4462$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
15	17	-3.25	5.125	-16.65625
17	13	-1.25	1.125	-1.40625
25	5	6.75	-6.875	-46.40625
27	7	8.75	-4.875	-42.65625
17	7	-1.25	-4.875	6.09375
12	21	-6.25	9.125	-57.03125
11	19	-7.25	7.125	-51.65625
22	6	3.75	-5.875	-22.03125
			Total	**-231.75**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-231.75}{(8-1)(5.8737)(6.4462)} \\ &\approx -0.8744 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.8744\frac{6.4462}{5.8737} = -0.9596$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 11.875 - (-0.9596) \cdot 18.25 = 29.3882$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 29.3882 - 0.9596x$$

Step 4

(b) Evaluate the regression equation found in part (a) at $x = 20$:

$$\hat{y} = a + bx = 29.3882 - 0.9596(20) = 10.1957$$

Thus the estimate number of crimes for the city is 10.1957 crimes.

Step 5

(c) The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.8744\frac{6.4462}{5.8737} = -0.9596$$

The slope represents the average increase (or decreased when the slope is negative) of the y-variable per unit of the x-variable.

The number of crimes decreases on average by 0.9596 crimes per police officer.

Result

(a) $\hat{y} = 29.3882 - 0.9596x$

(b) $\hat{y} = 10.1957$ crimes

(c) The number of crimes decreases on average by 0.9596 crimes per police officer.

PROBLEM STATEMENT

Chapter 13, Page 483

Step 1

Given:

$$n = \text{Sample size} = 12$$

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{9 + 7 + 11 + ... + 6}{12} = 8.9167$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{8.1 + 6 + 3.6 + ... + 6}{12} = 7.475$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(9 - 8.9167)^2 + ... + (6 - 8.9167)^2}{12 - 1}} \approx 2.2344$$
$$s_y = \sqrt{\frac{(8.1 - 7.475)^2 + ... + (6 - 7.475)^2}{12 - 1}} \approx 2.7971$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
9	8.1	0.0833	0.625	0.0521
7	6	-1.9167	-1.475	2.8271
11	3.6	2.0833	-3.875	-8.0728
12	4	3.0833	-3.475	-10.7145
8	5	-0.9167	-2.475	2.2688
7	10	-1.9167	2.525	-4.8397
8	8	-0.9167	0.525	-0.4813
11	11	2.0833	3.525	7.3436
10	10	1.0833	2.525	2.7353
12	12	3.0833	4.525	13.9519
6	6	-2.9167	-1.475	4.3021
6	6	-2.9167	-1.475	4.3021
			Total	**13.675**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{13.675}{(12-1)(2.2344)(6.1237)} \\ &\approx 0.1989 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.1989\frac{6.1237}{2.2344} = 0.2490$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 7.4750 - 0.2490 \cdot 8.9167 = 5.2546$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 5.2546 + 0.2490x$$

Step 4

(b) Evaluate the regression equation found in part (a) at $x = 10$:

$$\hat{y} = a + bx = 5.2546 + 0.2490(10) = 7.7448$$

Thus the estimated selling price of a 10-ear-old car is 7.7448 thousand dollars.

Step 5

(c) The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.1989\frac{6.1237}{2.2344} = 0.2490$$

The slope represents the average increase (or decreased when the slope is negative) of the y-variable per unit of the x-variable.

The selling price increases on average by 0.2490 thousand dollars per year.

🎯 Result

(a) $\hat{y} = 5.2546 + 0.2490x$

(b) $\hat{y} = 7.7448$ thousand dollars

(c) The selling price increases on average by 0.2490 thousand dollars per year.

PROBLEM STATEMENT

Chapter 13, Page 486

Step 1

Given:

$$n = \text{Sample size} = 8$$
$$\hat{y} = 29.29 - 0.96x$$
$$s_b = 0.22$$
$$\alpha = \text{Significance level} = 0.05$$

The estimate of the slope b is the coefficient of x in the given regression equation:

$$b = -0.96$$

Given claim: Slope is less than zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 \geq 0$$

$$H_1 : \beta_1 < 0$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 8 - 2 = 6$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.943$$

The rejection region then contains all values smaller than -1.943 and thus we reject H_0 when the t-value is smaller than -1.943.

Compute the value of the test statistic:

$$t = \frac{b - 0}{s_b} = \frac{-0.96 - 0}{0.22} \approx -4.3636$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -4.3636 < -1.943 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the slope of the regression line is less than zero.

Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 486

Step 1

Given:

$$n = \text{Sample size} = 12$$
$$\hat{y} = 11.18 - 0.49x$$
$$s_b = 0.23$$
$$\alpha = \text{Significance level} = 0.05$$

The estimate of the slope b is the coefficient of x in the given regression equation:

$$b = -0.49$$

Given claim: Slope is less than zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 \geq 0$$

$$H_1 : \beta_1 < 0$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 12 - 2 = 10$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.812$$

The rejection region then contains all values smaller than -1.812 and thus we reject H_0 when the t-value is smaller than -1.812.

Compute the value of the test statistic:

$$t = \frac{b - 0}{s_b} = \frac{-0.49 - 0}{0.23} \approx -2.1304$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -2.1304 < -1.812 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the slope of the regression line is less than zero.

Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 486

Step 1

Given:

$$n = \text{Sample size} = 12$$
$$\hat{y} = 1.85 + 0.08x$$
$$s_b = 0.03$$
$$\alpha = \text{Significance level} = 0.05$$

The estimate of the slope b is the coefficient of x in the given regression equation:

$$b = 0.08$$

Given claim: Slope is different from zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 12 - 2 = 10$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.228$$

The rejection region then contains all values smaller than -2.228 and all values greater than 2.228, thus we reject H_0 when $t < -2.228$ or $t > 2.228$.

Compute the value of the test statistic:

$$t = \frac{b - 0}{s_b} = \frac{0.08 - 0}{0.03} \approx 2.667$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 2.667 > 2.228 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the slope of the regression line is different from zero.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 486

Solution A

Step 1

We are already given the information necessary to start testing a hypothesis. These are:

Regression Equation: $\hat{y} = 9.9198 - 0.00039x$

Sample Size: $n = 9$

Standard Error: $s_b = 0.0032$

Significance Level: $\alpha = 0.05$

Using these values we may conduct hypothesis testing to know if the slope of the regression line is less than zero. Proceed to perform the 6 steps of hypothesis testing:

Step 1: Stating the hypotheses.

Step 2: Determining the level of significance.

Step 3: Determining the test statistic.

Step 4: Identifying the decision rule.

Step 5: Making a decision.

Step 6: Interpreting the results

Step 2

Testing of Hypothesis:

Step 1: Null and Alternate Hypothesis

Our goal is to determine whether the slope of the regression line is less than zero. Thus, our **null hypothesis** is that the population's slope is greater than or equal to 0 while the **alternate hypothesis** is that the population's slope is less than 0. We can express these hypotheses as:

$$H_0 : \ \beta \geq 0$$
$$H_1 : \ \beta < 0$$

Here, we will be using a *one-tailed test* because of the inequality. And since it is less than, then the region of rejection is in the *left tail* of the distribution.

Step 3

Step 2: Level of Significance

Now, we determine the level of significance. Based on the given information, we will be using 0.05 level of significance:

$$\alpha = 0.05$$

Step 4

Step 3: Test Statistic

Recall that we wanted to know if the slope of the regression line in the population is less than zero. To test the null hypothesis for the slope of the regression line, we will use the $t-$distribution with $n-2$ degrees of freedom. The value of test-statistic can be calculated by:

$$t = \frac{b - 0}{s_b}$$

where b is the estimated slope, and s_b is the standard error of the slope estimate.

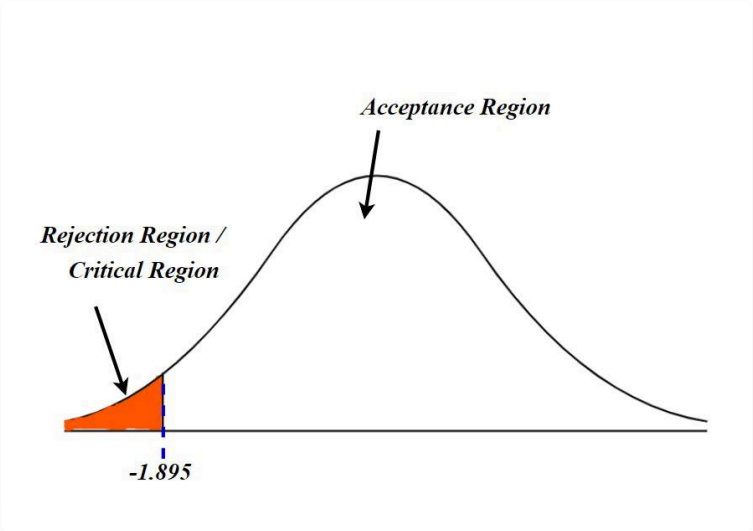
Step 5

Step 4: Decision Rule

To formulate the decision rule, we need to use the table for the critical values of t for the one-tailed test. To locate the critical value t, we first determine the degrees of freedom (df). This value is the difference between the total number of observations and 2. In the information given, there are 9 samples. Thus,

$$df = 9 - 2 = 7$$

The value on column 0.05 of one-tailed level of confidence and row 7 is 1.895 but because our rejection region is on the left side of the distribution, then we will take the negative of the value. Hence, our critical region is -1.895. Therefore, **the decision rule is to reject H_0 if the value of t is less than -1.895**.



Step 6

Step 5: Decision

Now that we have a decision rule, we proceed to calculate t. We know that our $n = 9$, and the standard error of slope estimate is 0.0032. We do not know the estimated slope yet. To find the slope of the regression equation, we will be looking at the equation itself. The given equation corresponds to the standard form of a linear equation:

$$\hat{y} = mx + b \longrightarrow \hat{y} = -0.00039x + 9.9198$$

thus, $m = -0.00039$. m indicates the slope of the regression line thus, we will be using $b = -0.00039$.

We can now proceed to calculating t by substituting $b = -0.00039$, $s_b = 0.0032$ into the t formula:

$$\begin{aligned} t &= \frac{b - 0}{s_b} \\ &= \frac{-0.00039 - 0}{0.0032} \\ &= \frac{-0.00039}{0.0032} \\ &= \frac{-0.00039}{0.0032} \\ &\approx -0.1219 \end{aligned}$$

Since the computed t value is -0.1219, then we fail to reject the null hypothesis because -0.1219 is greater than the critical value -1.895 and -0.1219 falls into the acceptance region.

$$\text{Decision: } t = -0.1219 > -1.895 \longrightarrow \text{Fail to reject } H_0$$

Step 7

Step 6: Interpretation

We fail to reject the null hypothesis which says that the population's slope is at least 0. This means that **we cannot conclude that the slope of the regression line is less than zero**.

Step 1

Given:

$$n = \text{Sample size} = 9$$
$$\hat{y} = 9.9198 - 0.00039x$$
$$s_b = 0.0032$$
$$\alpha = \text{Significance level} = 0.05$$

The estimate of the slope b is the coefficient of x in the given regression equation:

$$b = -0.00039$$

Given claim: Slope is less than zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 \geq 0$$

$$H_1 : \beta_1 < 0$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 9 - 2 = 7$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -2.365$$

The rejection region then contains all values smaller than -2.365 and thus we reject H_0 when the t-value is smaller than -2.365.

Compute the value of the test statistic:

$$t = \frac{b - 0}{s_b} = \frac{-0.00039 - 0}{0.0032} \approx -0.1219$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -0.1219 > -2.365 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the slope of the regression line is less than zero.

Result

No

Solution B

Step 1

We are already given the information necessary to start testing a hypothesis. These are:

Regression Equation: $\hat{y} = 9.9198 - 0.00039x$

Sample Size: $n = 9$

Standard Error: $s_b = 0.0032$

Significance Level: $\alpha = 0.05$

Using these values we may conduct hypothesis testing to know if the slope of the regression line is less than zero. Proceed to perform the 6 steps of hypothesis testing:

Step 1: Stating the hypotheses.

Step 2: Determining the level of significance.

Step 3: Determining the test statistic.

Step 4: Identifying the decision rule.

Step 5: Making a decision.

Step 6: Interpreting the results

Step 2

Testing of Hypothesis:

Step 1: Null and Alternate Hypothesis

Our goal is to determine whether the slope of the regression line is less than zero. Thus, our **null hypothesis** is that the population's slope is greater than or equal to 0 while the **alternate hypothesis** is that the population's slope is less than 0. We can express these hypotheses as:

$$H_0 : \ \beta \geq 0$$
$$H_1 : \ \beta < 0$$

Here, we will be using a *one-tailed test* because of the inequality. And since it is less than, then the region of rejection is in the *left tail* of the distribution.

Step 3

Step 2: Level of Significance

Now, we determine the level of significance. Based on the given information, we will be using 0.05 level of significance:

$$\alpha = 0.05$$

Step 4

Step 3: Test Statistic

Recall that we wanted to know if the slope of the regression line in the population is less than zero. To test the null hypothesis for the slope of the regression line, we will use the $t-$distribution with $n-2$ degrees of freedom. The value of test-statistic can be calculated by:

$$t = \frac{b - 0}{s_b}$$

where b is the estimated slope, and s_b is the standard error of the slope estimate.

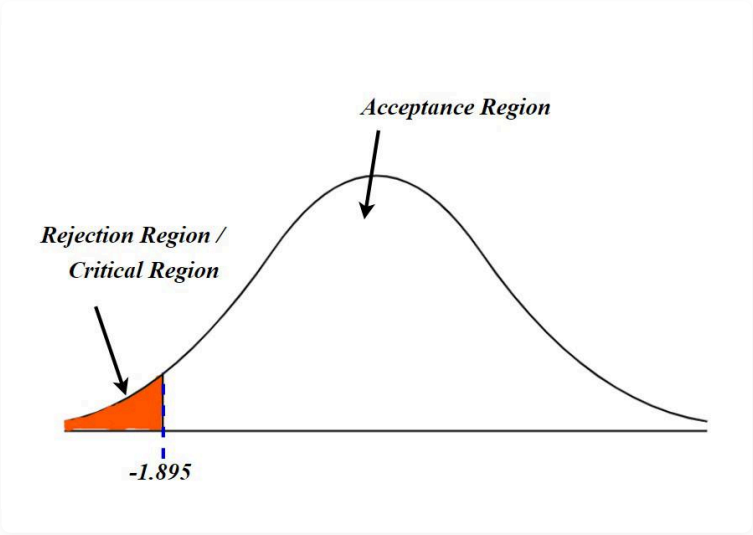
Step 5

Step 4: Decision Rule

To formulate the decision rule, we need to use the table for the critical values of t for the one-tailed test. To locate the critical value t, we first determine the degrees of freedom (df). This value is the difference between the total number of observations and 2. In the information given, there are 9 samples. Thus,

$$df = 9 - 2 = 7$$

The value on column 0.05 of one-tailed level of confidence and row 7 is 1.895 but because our rejection region is on the left side of the distribution, then we will take the negative of the value. Hence, our critical region is -1.895. Therefore, **the decision rule is to reject H_0 if the value of t is less than -1.895**.



Step 6

Step 5: Decision

Now that we have a decision rule, we proceed to calculate t. We know that our $n = 9$, and the standard error of slope estimate is 0.0032. We do not know the estimated slope yet. To find the slope of the regression equation, we will be looking at the equation itself. The given equation corresponds to the standard form of a linear equation:

$$\hat{y} = mx + b \longrightarrow \hat{y} = -0.00039x + 9.9198$$

thus, $m = -0.00039$. m indicates the slope of the regression line thus, we will be using $b = -0.00039$.

We can now proceed to calculating t by substituting $b = -0.00039$, $s_b = 0.0032$ into the t formula:

$$\begin{aligned} t &= \frac{b - 0}{s_b} \\ &= \frac{-0.00039 - 0}{0.0032} \\ &= \frac{-0.00039}{0.0032} \\ &= \frac{-0.00039}{0.0032} \\ &\approx -0.1219 \end{aligned}$$

Since the computed t value is -0.1219, then we fail to reject the null hypothesis because -0.1219 is greater than the critical value -1.895 and -0.1219 falls into the acceptance region.

$$\text{Decision: } t = -0.1219 > -1.895 \longrightarrow \text{Fail to reject } H_0$$

Step 7

Step 6: Interpretation

We fail to reject the null hypothesis which says that the population's slope is at least 0. This means that **we cannot conclude that the slope of the regression line is less than zero**.

Step 1

Given:

$$n = \text{Sample size} = 9$$
$$\hat{y} = 9.9198 - 0.00039x$$
$$s_b = 0.0032$$
$$\alpha = \text{Significance level} = 0.05$$

The estimate of the slope b is the coefficient of x in the given regression equation:

$$b = -0.00039$$

Given claim: Slope is less than zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 \geq 0$$

$$H_1 : \beta_1 < 0$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 9 - 2 = 7$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -2.365$$

The rejection region then contains all values smaller than -2.365 and thus we reject H_0 when the t-value is smaller than -2.365.

Compute the value of the test statistic:

$$t = \frac{b - 0}{s_b} = \frac{-0.00039 - 0}{0.0032} \approx -0.1219$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -0.1219 > -2.365 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the slope of the regression line is less than zero.

Result

No

PROBLEM STATEMENT

Chapter 13, Page 488

Step 1

Given:

$$n = \text{Sample size} = 8$$

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{15 + 17 + 25 + 17 + 17 + 12 + 11 + 22}{8} = 18.25$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{17 + 13 + 5 + 7 + 7 + 21 + 19 + 6}{8} = 11.875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(15 - 18.25)^2 + ... + (22 - 18.25)^2}{8 - 1}} \approx 5.8737$$
$$s_y = \sqrt{\frac{(17 - 11.875)^2 + ... + (6 - 11.875)^2}{8 - 1}} \approx 6.4462$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
15	17	-3.25	5.125	-16.65625
17	13	-1.25	1.125	-1.40625
25	5	6.75	-6.875	-46.40625
27	7	8.75	-4.875	-42.65625
17	7	-1.25	-4.875	6.09375
12	21	-6.25	9.125	-57.03125
11	19	-7.25	7.125	-51.65625
22	6	3.75	-5.875	-22.03125
			Total	-231.75

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-231.75}{(8-1)(5.8737)(6.4462)} \\ &\approx -0.8744 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.8744\frac{6.4462}{5.8737} = -0.9596$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 11.875 - (-0.9596) \cdot 18.25 = 29.3882$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 29.3882 - 0.9596x$$

Step 4

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y-\hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y-\hat{Y}$	$(Y-\hat{Y})^2$
15	17	14.9942	2.0058	4.0232
17	13	13.075	-0.075	0.0056
25	5	5.3982	-0.3982	0.1586
27	7	3.479	3.521	12.3974
17	7	13.075	-6.075	36.9056
12	21	17.873	3.127	9.7781
11	19	18.8326	0.1674	0.0280
22	6	8.2770	-2.770	5.1847
			Total	68.4814

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n-2$.

$$s=\sqrt{\frac{\sum(Y-\hat{Y})^2}{n-2}}=\sqrt{\frac{68.4814}{8-2}}\approx 3.3784$$

Step 5

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2=(-0.8744)^2\approx 0.7646=76.46\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

76.46% of the variation in the number of crimes is explained by the variation in the number of police officers.

Result

Standard error of the estimate: $s=3.3784$

Coefficient of determination: 76.46% of the variation in the number of crimes is explained by the variation in the number of police officers.

PROBLEM STATEMENT

Chapter 13, Page 488

Step 1

Given:

$$n = \text{Sample size} = 12$$

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{9 + 7 + 11 + ... + 6}{12} = 8.9167$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{8.1 + 6 + 3.6 + ... + 6}{12} = 7.475$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(9 - 8.9167)^2 + ... + (6 - 8.9167)^2}{12 - 1}} \approx 2.2344$$
$$s_y = \sqrt{\frac{(8.1 - 7.475)^2 + ... + (6 - 7.475)^2}{12 - 1}} \approx 2.7971$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
9	8.1	0.0833	0.625	0.0521
7	6	-1.9167	-1.475	2.8271
11	3.6	2.0833	-3.875	-8.0728
12	4	3.0833	-3.475	-10.7145
8	5	-0.9167	-2.475	2.2688
7	10	-1.9167	2.525	-4.8397
8	8	-0.9167	0.525	-0.4813
11	11	2.0833	3.525	7.3436
10	10	1.0833	2.525	2.7353
12	12	3.0833	4.525	13.9519
6	6	-2.9167	-1.475	4.3021
6	6	-2.9167	-1.475	4.3021
			Total	13.675

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{13.675}{(12-1)(2.2344)(6.1237)} \\ &\approx 0.1989 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.1989\frac{6.1237}{2.2344} = 0.2490$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 7.4750 - 0.2490 \cdot 8.9167 = 5.2546$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 5.2546 + 0.2490x$$

Step 4

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y-\hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y-\hat{Y}$	$(Y-\hat{Y})^2$
9	8.1	7.4956	0.6044	0.3653
7	6	6.9976	-0.9976	0.9952
11	3.6	7.9936	-4.3936	19.3037
12	4	8.2426	-4.2426	17.9997
8	5	7.2466	-2.2466	5.0472
7	10	6.9976	3.0024	9.0144
8	8	7.2466	0.7534	0.5676
11	11	7.9936	3.0064	9.0384
10	10	7.7446	2.2554	5.0868
12	12	8.2426	3.7574	14.1181
6	6	6.7486	-0.7486	0.5604
6	6	6.7486	-0.7486	0.5604
			Total	82.6572

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n-2$.

$$s=\sqrt{\frac{\sum(Y-\hat{Y})^2}{n-2}}=\sqrt{\frac{82.6572}{12-2}}\approx 2.8750$$

Step 5

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2=(0.1989)^2\approx 0.0396=3.96\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

3.96% of the variation in the selling price is explained by the variation in the age of the car.

Result

Standard error of the estimate: $s=2.8750$

Coefficient of determination: 3.96% of the variation in the selling price is explained by the variation in the age of the car.

PROBLEM STATEMENT

Chapter 13, Page 488

Step 1

Given:

$$n = \text{Sample size} = 10$$

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{12 + 9 + 14 + ... + 7}{10} = 9.1$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{9 + 7 + 10 + ... + 7}{10} = 7.4$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(12 - 9.1)^2 + ... + (7 - 9.1)^2}{10 - 1}} \approx 2.7264$$
$$s_y = \sqrt{\frac{(9 - 7.4)^2 + ... + (7 - 7.4)^2}{10 - 1}} \approx 2.0111$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
12	9	2.9	1.6	4.64
9	7	-0.1	-0.4	0.04
14	10	4.9	2.6	12.74
6	5	-3.1	-2.4	7.44
10	8	0.9	0.6	0.54
8	6	-1.1	-1.4	1.54
10	8	0.9	0.6	0.54
10	10	0.9	2.6	2.34
5	4	-4.1	-3.4	13.94
7	7	-2.1	-0.4	0.84
			Total	44.6

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1) s_x s_y} \\ &= \frac{44.6}{(10-1)(2.7264)(2.0111)} \\ &\approx 0.9038 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r \frac{s_y}{s_x} = 0.9038 \frac{2.0111}{2.7264} = 0.6667$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 7.4 - 0.6667 \cdot 9.1 = 1.3333$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 1.3333 + 0.6667x$$

Step 4

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y-\hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y-\hat{Y}$	$(Y-\hat{Y})^2$
12	9	9.3337	-0.3337	0.1114
9	7	7.3336	-0.3336	0.1113
14	10	10.6771	-0.6671	0.4450
6	5	5.3335	-0.3335	0.1112
10	8	8.0003	-0.0003	0.0000
8	6	6.6669	-0.6669	0.4448
10	8	8.0003	-0.0003	0.0000
10	10	8.0003	1.9997	3.9988
5	4	4.6668	-0.6668	0.4446
7	7	6.0002	0.9998	0.9996
			Total	6.6667

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n-2$.

$$s=\sqrt{\frac{\sum(Y-\hat{Y})^2}{n-2}}=\sqrt{\frac{6.6667}{10-2}}\approx 0.9129$$

Step 5

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2=(0.9038)^2\approx 0.8168=81.68\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

81.68% of the variation in the kilowatt-hours is explained by the variation in the number of rooms

Result

Standard error of the estimate: $s=0.9129$

Coefficient of determination: 81.68% of the variation in the kilowatt-hours is explained by the variation in the number of rooms

PROBLEM STATEMENT

Chapter 13, Page 488

Step 1

Given:

$$n = \text{Sample size} = 10$$

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{14 + 12 + 20 + ... + 50}{10} = 33.4$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{24 + 14 + 28 + ... + 110}{10} = 61.1$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(14 - 33.4)^2 + ... + (50 - 33.4)^2}{10 - 1}} \approx 17.6836$$
$$s_y = \sqrt{\frac{(24 - 61.1)^2 + ... + (110 - 61.1)^2}{10 - 1}} \approx 39.7896$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
14	24	-19.4	-37.1	719.74
12	14	-21.4	-47.1	1007.94
20	28	-13.4	-33.1	443.54
16	30	-17.4	-31.1	541.14
46	80	12.6	18.9	238.14
23	30	-10.4	-31.1	323.44
48	90	14.6	28.9	421.94
50	85	16.6	23.9	396.74
55	120	21.6	58.9	1272.24
50	110	16.6	48.9	81.74
			Total	6176.6

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{6176.6}{(10-1)(17.6836)(39.7896)} \\ &\approx 0.9754 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9754\frac{39.7896}{17.6836} = 2.1946$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 61.1 - 2.1946 \cdot 33.4 = -12.2010$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -12.2010 + 2.1946x$$

Step 4

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y-\hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y-\hat{Y}$	$(Y-\hat{Y})^2$
14	24	18.5234	5.4766	29.9931
12	14	14.1342	-0.1342	0.0180
20	28	31.6910	-3.6910	13.6235
16	30	22.9126	7.0874	50.2312
46	80	88.7506	-8.7506	76.5730
23	30	38.2748	-8.2748	68.4723
48	90	93.1398	-3.1398	9.8583
50	85	97.5290	-12.5290	156.9758
55	120	108.5020	11.4980	132.2040
50	110	97.5290	12.4710	155.5258
			Total	693.4752

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n-2$.

$$s=\sqrt{\frac{\sum(Y-\hat{Y})^2}{n-2}}=\sqrt{\frac{693.4752}{10-2}}\approx 9.3104$$

Step 5

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2=0.9754^2\approx 0.9513=95.13\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

95.13% of the variation in the sales is explained by the variation in the number of contacts.

Result

Standard error of the estimate: $s=9.3104$

Coefficient of determination: 95.13% of the variation in the sales is explained by the variation in the number of contacts.

PROBLEM STATEMENT

Chapter 13, Page 490

Step 1

(a) The regression sum of squares is given in the row "Regression" and in the column "SS" of the given ANOVA table:

$$SSR = 1000.0$$

The total sum of squares is given in the row "Total" and in the column "SS" of the given ANOVA table:

$$\text{Total } SS = 1500.0$$

The coefficient of determination is SSR divided by Total SS:

$$r^2 = \frac{SSR}{\text{Total } SS} = \frac{1000.0}{1500.0} = \frac{2}{3} \approx 0.6677 = 67.77\%$$

Step 2

(b) The correlation coefficient is the positive or negative square root of the coefficient of determination.

We have been given that there is a direct (positive) relationship between the variables, which implies that the correlation coefficient is positive.

The correlation coefficient is then the positive square root of the coefficient of determination:

$$r = +\sqrt{r^2} = +\sqrt{0.6677} \approx 0.8171$$

Step 3

(c) The error sum of squares is given in the row "Error" and in the column "SS" of the given ANOVA table:

$$SSE = 500.0$$

The error degrees of freedom is given in the row "Error" and in the column "DF" of the given ANOVA table:

$$d.f._E = 13$$

The variance s^2 is the error sum of squares divided by the error degrees of freedom. The standard error of the estimate s is the square root of the variance:

$$s = \sqrt{\frac{SSE}{d.f._E}} = \sqrt{\frac{500.0}{13}} \approx 6.2017$$

🎯 **Result**

(a) $r^2 = \frac{2}{3} \approx 0.6677 = 67.77\%$

(b) $r = 0.8171$

(c) $s = 6.2017$

PROBLEM STATEMENT

Chapter 13, Page 490

Step 1

Give:

$$r^2 = \text{Coefficient of determination} = 80\% = 0.80$$
$$s = \text{Standard error of the estimate} = 10$$
$$n = \text{Sample size} = 20$$

The regression degrees of freedom is equal to 1 for a simple linear regression with one prediction variable:

$$d.f._R = 1$$

The error degrees of freedom is the sample size n decreased by 2.

$$d.f._E = n - 2 = 20 - 2 = 18$$

The total degrees of freedom is the sum of the regression degrees of freedom and the error degrees of freedom.

$$d.f._{total} = d.f._R + d.f._E = 1 + 18 = 19$$

Step 2

The variance s^2 is the error sum of squares divided by the error degrees of freedom. The standard error of the estimate s is the square root of the variance:

$$s = \sqrt{\frac{SSE}{d.f._E}} \Rightarrow SSE = s^2 \times d.f._E = 10^2 \times 18 = 1800$$

The coefficient of determination is SSR divided by Total SS. Since the coefficient of determination is 80% and Total $SS = SSR + SSE$, the SSE represents $100\% - 80\% = 20\%$ of the Total SS and thus the Total SS is five times the SSE.

$$\text{Total } SS = \frac{SSE}{1 - r^2} = \frac{1800}{1 - 0.80} = \frac{1800}{0.20} = 1800 \times 5 = 9000$$

Since the equality Total $SS = SSR + SSE$ holds:

$$SSR = \text{Total } SS - SSE = 9000 - 1800 = 7200$$

Step 3

The MSR is the SSR divided by $d.f._R$:

$$MSR = \frac{SSR}{d.f._R} = \frac{7200}{1} = 7200$$

The MSE is the SSE divided by $d.f._E$:

$$MSE = \frac{SSE}{d.f._E} = \frac{1800}{18} = 100$$

The value of the F test statistic is the ratio of the MSR and the MSE.

$$F = \frac{MSR}{MSE} = \frac{7200}{100} = 72$$

Let us combine the information in an ANOVA table:

Source	df	SS	MS	F
Regression	1	7200	7200	72
Error	18	1800	100	
Total	19	9000		

Result

Source	df	SS	MS	F
Regression	1	7200	7200	72
Error	18	1800	100	
Total	19	9000		

PROBLEM STATEMENT

Chapter 13, Page 494

Step 1

Given:

$$n = \text{Sample size} = 5$$
$$c = \text{Confidence interval} = 95\%$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{4+5+3+6+10}{5} = 5.6$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{4+6+5+7+7}{5} = 5.8$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(4-5.6)^2 + ... + (10-5.6)^2}{5-1}} \approx 2.7019$$
$$s_y = \sqrt{\frac{(4-5.8)^2 + ... + (7-5.8)^2}{5-1}} \approx 1.3038$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$	$(X - \overline{X})^2$
4	4	-1.6	-1.8	2.88	203.0625
5	6	-0.6	0.2	-0.12	175.5625
3	5	-2.6	-0.8	2.08	232.5625
6	7	0.4	1.2	0.48	150.0625
10	7	4.4	1.2	5.28	68.0625
			Total	10.6	829.3125

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{10.6}{(5-1)(2.7019)(1.3038)} \\ &\approx 0.7522 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.8744\frac{6.4462}{5.8737} = -0.9596$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 11.875 - (-0.9596) \cdot 18.25 = 29.3882$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 29.3882 - 0.9596x$$

Step 4

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y-\hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y-\hat{Y}$	$(Y-\hat{Y})^2$
4	4	5.2191	-1.2191	1.4862
5	6	5.5821	0.4179	0.1746
3	5	4.8561	0.1439	0.0207
6	7	5.9451	1.0549	1.1128
10	7	7.3971	-0.3971	0.1577
			Total	2.9521

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n-2$.

$$s=\sqrt{\frac{\sum(Y-\hat{Y})^2}{n-2}}=\sqrt{\frac{2.9521}{5-2}}\approx 0.9920$$

Step 5

Confidence interval for the mean

Evaluate the regression equation at $x=x_0=7$:

$$\hat{y}=a+bx=3.7671+0.3630(7)=6.3082$$

Determine the critical value using the Student's T distribution table in the appendix with $df=n-2=5-2=3$ and $c=95\%$:

$$t_c=\pm 3.182$$

Determine the margin of error:

$$\begin{aligned}E&=t_c s\sqrt{\frac{1}{n}+\frac{(x_0-\overline{x})^2}{\sum(x-\overline{x})^2}}\\&=1.895(0.9920)\sqrt{\frac{1}{5}+\frac{(7-5.6)^2}{829.3125}}\\&\approx 0.8456\end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned}\hat{y}-E&=6.3082-0.8456=5.4626\\\hat{y}+E&=6.3082+0.8456=7.1538\end{aligned}$$

Step 6

(b) **Prediction interval for an individual**

Evaluate the regression equation at $x = x_0 = 7$:

$$\hat{y} = a + bx = 3.7671 + 0.3630(7) = 6.3082$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 5 - 2 = 3$ and $c = 95\%$:

$$t_c = \pm 3.182$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 1.895(0.9920)\sqrt{1 + \frac{1}{5} + \frac{(7 - 5.6)^2}{829.3125}} \\ &\approx 2.0613 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 6.3082 - 2.0613 = 4.2469 \\ \hat{y} + E &= 6.3082 + 2.0613 = 8.3695 \end{aligned}$$

Result

(a) $(5.4626, 7.1538)$

(b) $(4.2469, 8.3695)$

PROBLEM STATEMENT

Chapter 13, Page 494

Step 1

Given:

$$n = \text{Sample size} = 8$$
$$c = \text{Confidence interval} = 95\%$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{5 + 3 + 6 + ... + 8}{8} = 4.875$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{13 + 15 + 7 + ... + 5}{8} = 10.625$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(5 - 4.875)^2 + ... + (8 - 4.875)^2}{8 - 1}} \approx 1.7269$$
$$s_y = \sqrt{\frac{(13 - 10.625)^2 + ... + (5 - 10.625)^2}{8 - 1}} \approx 3.3780$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$	$(X - \overline{X})^2$
5	13	0.125	2.375	0.296875	0.0156
3	15	-1.875	4.375	-8.203125	3.5156
6	7	1.125	-3.625	-4.078125	1.2656
3	12	-1.875	1.375	-2.57812	3.5156
4	13	-0.875	2.375	-2.078125	0.7656
4	11	-0.875	0.375	-0.328125	0.7656
6	9	1.125	-1.625	-1.828125	1.2656
8	5	3.125	-5.625	-17.5781	9.7656
			Total	-36.375	20.875

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-36.375}{(8-1)(1.7269)(3.3780)} \\ &\approx -0.8908 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.8908\frac{3.3780}{1.7269} = -1.7425$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 10.625 - (-1.7425) \cdot 4.875 = 19.1198$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 19.1198 - 1.7425x$$

Step 4

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y-\hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y-\hat{Y}$	$(Y-\hat{Y})^2$
5	13	10.4073	2.5927	6.7221
3	15	13.8923	1.1077	1.2270
6	7	8.6648	-1.6648	2.7716
3	12	13.8923	-1.8923	3.5808
4	13	12.1498	0.8502	0.7228
4	11	12.1498	-1.1498	1.3220
6	9	8.6648	0.3352	0.1124
8	5	5.1798	-0.1798	0.0323
			Total	16.4910

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n-2$.

$$s=\sqrt{\frac{\sum(Y-\hat{Y})^2}{n-2}}=\sqrt{\frac{16.4910}{8-2}}\approx 1.6579$$

Step 5

Confidence interval for the mean

Evaluate the regression equation at $x=x_0=7$:

$$\hat{y}=a+bx=19.1198-1.7425(7)=6.9222$$

Determine the critical value using the Student's T distribution table in the appendix with $df=n-2=8-2=6$ and $c=95\%$:

$$t_c=\pm 2.447$$

Determine the margin of error:

$$\begin{aligned}E&=t_c s\sqrt{\frac{1}{n}+\frac{(x_0-\overline{x})^2}{\sum(x-\overline{x})^2}}\\&=2.447(1.6579)\sqrt{\frac{1}{8}+\frac{(7-4.875)^2}{20.875}}\\&\approx 2.3701\end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned}\hat{y}-E&=6.9222-2.3701=4.5521\\\hat{y}+E&=6.9222+2.3701=9.2923\end{aligned}$$

Step 6

(b) **Prediction interval for an individual**

Evaluate the regression equation at $x = x_0 = 7$:

$$\hat{y} = a + bx = 19.1198 - 1.7425(7) = 6.9222$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 8 - 2 = 6$ and $c = 95\%$:

$$t_c = \pm 2.447$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 2.447(1.6579)\sqrt{1 + \frac{1}{8} + \frac{(7 - 4.875)^2}{20.875}} \\ &\approx 4.6985 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 6.9222 - 4.6985 = 2.2237 \\ \hat{y} + E &= 6.9222 + 4.6985 = 11.6207 \end{aligned}$$

Result

(a) $(4.5521, 9.2923)$

(b) $(2.2237, 11.6207)$

PROBLEM STATEMENT

Chapter 13, Page 494

Step 1

Given:

$$n = \text{Sample size} = 10$$
$$c = \text{Confidence interval} = 95\%$$

(a)**Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{12 + 9 + 14 + ... + 7}{10} = 9.1$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{9 + 7 + 10 + ... + 7}{10} = 7.4$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(12 - 9.1)^2 + ... + (7 - 9.1)^2}{10 - 1}} \approx 2.7264$$
$$s_y = \sqrt{\frac{(9 - 7.4)^2 + ... + (7 - 7.4)^2}{10 - 1}} \approx 2.0111$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$	$(X - \overline{X})^2$
12	9	2.9	1.6	4.64	8.41
9	7	-0.1	-0.4	0.04	0.01
14	10	4.9	2.6	12.74	24.01
6	5	-3.1	-2.4	7.44	9.61
10	8	0.9	0.6	0.54	0.81
8	6	-1.1	-1.4	1.54	1.21
10	8	0.9	0.6	0.54	0.81
10	10	0.9	2.6	2.34	0.81
5	4	-4.1	-3.4	13.94	16.81
7	7	-2.1	-0.4	0.84	4.41
			Total	44.6	66.9

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{44.6}{(10-1)(2.7264)(2.0111)} \\ &\approx 0.9038 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9038\frac{2.0111}{2.7264} = 0.6667$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 7.4 - 0.6667 \cdot 9.1 = 1.3333$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 1.3333 + 0.6667x$$

Step 4

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y-\hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y-\hat{Y}$	$(Y-\hat{Y})^2$
12	9	9.3337	-0.3337	0.1114
9	7	7.3336	-0.3336	0.1113
14	10	10.6771	-0.6671	0.4450
6	5	5.3335	-0.3335	0.1112
10	8	8.0003	-0.0003	0.0000
8	6	6.6669	-0.6669	0.4448
10	8	8.0003	-0.0003	0.0000
10	10	8.0003	1.9997	3.9988
5	4	4.6668	-0.6668	0.4446
7	7	6.0002	0.9998	0.9996
			Total	6.6667

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n-2$.

$$s=\sqrt{\frac{\sum(Y-\hat{Y})^2}{n-2}}=\sqrt{\frac{6.6667}{10-2}}\approx 0.9129$$

Step 5

Confidence interval for the mean

Evaluate the regression equation found at $x = x_0 = 6$:

$$\hat{y} = a + bx = 1.3333 + 0.6667(6) = 5.3333$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 10 - 2 = 8$ and $c = 95\%$:

$$t_c = \pm 2.306$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{\frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 2.306(0.9129)\sqrt{\frac{1}{10} + \frac{(6 - 9.1)^2}{66.9}} \\ &\approx 1.0391 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 5.3333 - 1.0391 = 4.2942 \\ \hat{y} + E &= 5.3333 + 1.0391 = 6.3724 \end{aligned}$$

Step 6

(b) **Prediction interval for an individual**

Evaluate the regression equation found at $x = x_0 = 6$:

$$\hat{y} = a + bx = 1.3333 + 0.6667(6) = 5.3333$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 10 - 2 = 8$ and $c = 95\%$:

$$t_c = \pm 2.306$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 2.306(0.9129)\sqrt{1 + \frac{1}{10} + \frac{(6 - 9.1)^2}{66.9}} \\ &\approx 2.3476 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 5.3333 - 2.3476 = 2.9857 \\ \hat{y} + E &= 5.3333 + 2.3476 = 8.3695 \end{aligned}$$

Result

(a) $(4.2942, 6.3724)$

(b) $(2.9857, 7.6809)$

PROBLEM STATEMENT

Chapter 13, Page 495

Step 1

Given:

$$n = \text{Sample size} = 10$$
$$c = \text{Confidence interval} = 95\%$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{14 + 12 + 20 + ... + 50}{10} = 33.4$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{24 + 14 + 28 + ... + 110}{10} = 61.1$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(14 - 33.4)^2 + ... + (50 - 33.4)^2}{10 - 1}} \approx 17.6836$$
$$s_y = \sqrt{\frac{(24 - 61.1)^2 + ... + (110 - 61.1)^2}{10 - 1}} \approx 39.7896$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$	$(X - \overline{X})^2$
14	24	-19.4	-37.1	719.74	376.36
12	14	-21.4	-47.1	1007.94	457.96
20	28	-13.4	-33.1	443.54	179.56
16	30	-17.4	-31.1	541.14	302.76
46	80	12.6	18.9	238.14	158.76
23	30	-10.4	-31.1	323.44	108.16
48	90	14.6	28.9	421.94	213.16
50	85	16.6	23.9	396.74	275.56
55	120	21.6	58.9	1272.24	466.56
50	110	16.6	48.9	81.74	275.56
			Total	6176.6	2814.4

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{6176.6}{(10-1)(17.6836)(39.7896)} \\ &\approx 0.9754 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9754\frac{39.7896}{17.6836} = 2.1946$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 61.1 - 2.1946 \cdot 33.4 = -12.2010$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -12.2010 + 2.1946x$$

Step 4

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y - \hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y - \hat{Y}$	$(Y - \hat{Y})^2$
14	24	18.5234	5.4766	29.9931
12	14	14.1342	-0.1342	0.0180
20	28	31.6910	-3.6910	13.6235
16	30	22.9126	7.0874	50.2312
46	80	88.7506	-8.7506	76.5730
23	30	38.2748	-8.2748	68.4723
48	90	93.1398	-3.1398	9.8583
50	85	97.5290	-12.5290	156.9758
55	120	108.5020	11.4980	132.2040
50	110	97.5290	12.4710	155.5258
			Total	693.4752

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n - 2$.

$$s = \sqrt{\frac{\sum (Y - \hat{Y})^2}{n - 2}} = \sqrt{\frac{693.4752}{10 - 2}} \approx 9.3104$$

Step 5

Confidence interval for the mean

Evaluate the regression equation at $x = x_0 = 40$:

$$\hat{y} = a + bx = -12.2010 + 2.1946(40) = 75.5846$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 10 - 2 = 8$ and $c = 95\%$:

$$t_c = \pm 2.306$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{\frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 2.306(9.3104)\sqrt{\frac{1}{10} + \frac{(40 - 33.4)^2}{2814.4}} \\ &\approx 7.2959 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 75.5846 - 7.2959 = 68.2887 \\ \hat{y} + E &= 75.5846 + 7.2959 = 82.8805 \end{aligned}$$

Step 6

(b) **Prediction interval for an individual**

Evaluate the regression equation at $x = x_0 = 40$:

$$\hat{y} = a + bx = -12.2010 + 2.1946(40) = 75.5846$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 10 - 2 = 8$ and $c = 95\%$:

$$t_c = \pm 2.306$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 2.306(9.3104)\sqrt{1 + \frac{1}{10} + \frac{(40 - 33.4)^2}{2814.4}} \\ &\approx 22.6756 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 75.5846 - 22.6756 = 52.9090 \\ \hat{y} + E &= 75.5846 + 22.6756 = 98.2602 \end{aligned}$$

Result

(a) $(68.2887, 82.8805)$

(b) $(52.9090, 98.2602)$

PROBLEM STATEMENT

Chapter 13, Page 497

Step 1

Given:

$$n = \text{Sample size} = 5$$

Correlation coefficient regular data

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{-8 - 16 + 12 + 2 + 18}{5} = 1.6$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{58 + 247 + 153 + 3 + 341}{5} = 160.4$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(-8 - 1.6)^2 + ... + (18 - 1.6)^2}{5 - 1}} \approx 13.9571$$
$$s_y = \sqrt{\frac{(58 - 160.4)^2 + ... + (341 - 160.4)^2}{5 - 1}} \approx 137.3237$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X-\overline{X}$	$Y-\overline{Y}$	$(X-\overline{X})(Y-\overline{Y})$
-8	58	-9.6	-102.4	983.04
-16	247	-17.6	86.6	-1524.16
12	153	10.4	-7.4	-76.96
2	3	0.4	-157.4	-62.96
18	341	16.4	180.6	2961.84
			Total	2280.8

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{2280.8}{(5-1)(13.9571)(137.3237)} \\ &\approx 0.2975 \end{aligned}$$

We note that the correlation coefficient is close to 0, which indicates that the relationship between the variables does not appear to be linear.

Step 3

Correlation coefficient squared X-variable

Let X represent the square of the given X variable.

Formula correlation coefficient:

$$r = \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x} &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{64+256+144+4+324}{5} = 158.4 \\ \overline{y} &= \frac{\sum_{i=1}^{n} y_i}{n} = \frac{58+247+153+3+341}{5} = 160.4 \end{aligned}$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$\begin{aligned} s_x &= \sqrt{\frac{(64-158.4)^2+....+(324-158.4)^2}{5-1}} \approx 132.1998 \\ s_y &= \sqrt{\frac{(58-160.4)^2+....+(341-160.4)^2}{5-1}} \approx 137.3237 \end{aligned}$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
64	58	-94.4	-102.4	9666.56
256	247	97.6	86.6	8452.16
144	153	-14.4	-7.4	106.56
4	3	0.4	-154.4	24302.56
324	341	165.6	180.6	29907.36
			Total	72435.2

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{72435.2}{(5-1)(132.1998)(137.3237)} \\ &\approx 0.9975 \end{aligned}$$

Result

Regular data: $r = 0.2975$, not linear

Squared X-variable: $r = 0.9975$

PROBLEM STATEMENT

Chapter 13, Page 497

Step 1

Given:

$$n = \text{Sample size} = 10$$

(a) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2 + 5 + 8 + ... + 60}{10} = 23.1$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{120 + 90 + 80 + ... + 21}{10} = 56.2$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(2 - 23.1)^2 + ... + (60 - 23.1)^2}{10 - 1}} \approx 18.7880$$
$$s_y = \sqrt{\frac{(120 - 56.2)^2 + ... + (21 - 56.2)^2}{10 - 1}} \approx 32.7747$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
2	120	-21.1	63.8	-1346.18
5	90	-18.1	33.8	-611.78
8	80	-15.1	23.8	-359.38
12	70	-11.1	13.8	-153.18
16	50	-7.1	-6.2	44.02
21	45	-2.1	-11.2	23.52
27	31	3.9	-25.2	-98.28
35	30	11.9	-26.2	-311.78
45	25	21.9	-31.2	-683.28
60	21	36.9	-35.2	-1298.88
			Total	**-4795.2**

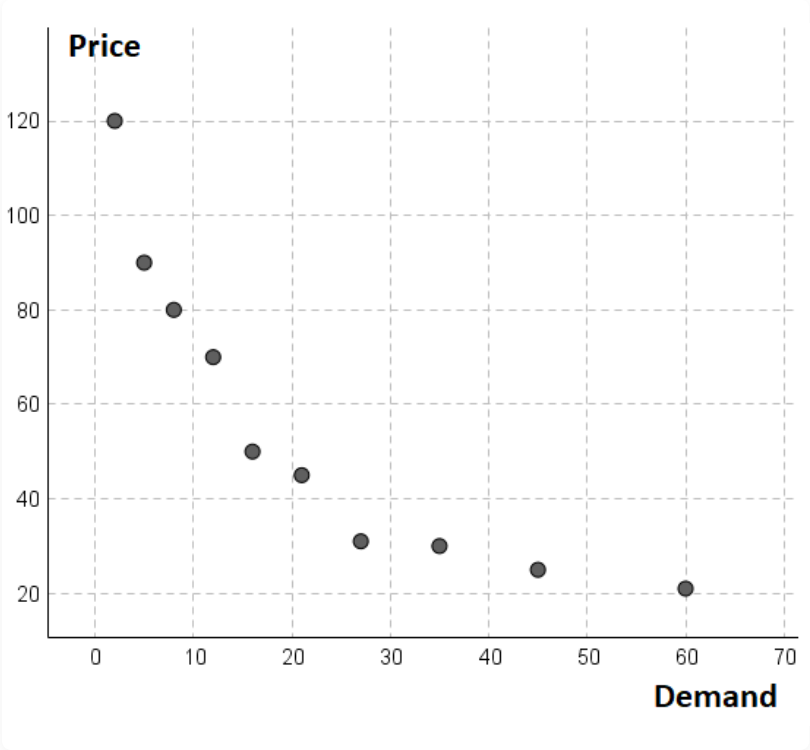
Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-4795.2}{(10-1)(18.7880)(32.7747)} \\ &\approx -0.8653 \end{aligned}$$

Step 3

Scatter diagram

Demand is on the horizontal axis and Price is on the vertical axis.



Step 4

We note that the scatter diagram contains strong curvature, which implies that the relationship is not linear.

Step 5

(b) Let Y represent the logarithm with base 10 of the price.

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2 + 5 + 8 + ... + 60}{10} = 23.1$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{2.0792 + 1.9542 + 1.9031 + ... + 1.3222}{10} = 1.6822$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(2 - 23.1)^2 + ... + (60 - 23.1)^2}{10 - 1}} \approx 18.7880$$
$$s_y = \sqrt{\frac{(2.0792 - 1.6822)^2 + ... + (1.3222 - 1.6822)^2}{10 - 1}} \approx 0.2575$$

Step 6

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
2	2.0792	-21.1	0.3970	-8.7363
5	1.9542	-18.1	0.2720	-4.9240
8	1.9031	-15.1	0.2209	-3.3354
12	1.8451	-11.1	0.1629	-1.8082
16	1.6990	-7.1	0.0168	-0.1191
21	1.6532	-2.1	-0.0290	0.0609
27	1.4914	3.9	-0.1908	-0.7443
35	1.4771	11.9	-0.2051	-2.4404
45	1.3979	21.9	-0.2843	-6.2253
60	1.3222	36.9	-0.3600	-13.2833
			Total	**-41.1954**

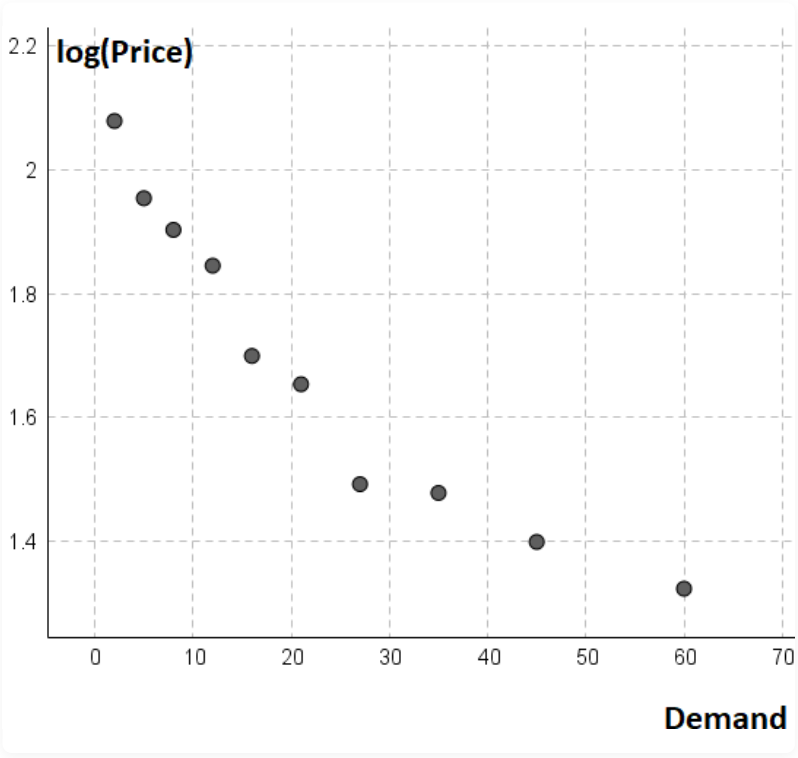
Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-41.1954}{(10-1)(18.7880)(0.2575)} \\ &\approx -0.9460 \end{aligned}$$

Step 7

Scatter diagram

Demand is on the horizontal axis and logarithm of Price is on the vertical axis.



Step 8

We note that this scatter diagram contains less curvature than the scatter diagram in part (a), which implies that the relationship has improved between the variables.

Result

(a) $r = -0.8653$, not linear

(b) $r = -0.9460$, improved

PROBLEM STATEMENT

Chapter 13, Page 500

Step 1

Given:

$$n = \text{Sample size} = 25$$
$$r = \text{Correlation coefficient} = 0.94$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 25 - 2 = 23$ and in the column with $\alpha = 0.05$.

$$t = 2.069$$

The rejection region then contains all values greater than 2.069 and thus we reject H_0 when the t-value is larger than 2.069.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.94\sqrt{25-2}}{\sqrt{1-0.94^2}} \approx 13.213$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$13.213 > 2.069 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a positive association between the two variables.

Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 500

Step 1

Given:

$$n = \text{Sample size} = 20$$
$$r = \text{Correlation coefficient} = 0.40$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 20 - 2 = 18$ and in the column with $\alpha = 0.01$.

$$t = 2.552$$

The rejection region then contains all values greater than 2.552 and thus we reject H_0 when the t-value is larger than 2.552.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.40\sqrt{20-2}}{\sqrt{1-0.40^2}} \approx 1.8516$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 1.8516 < 2.552 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is a positive correlation between the variables.

Result

No

PROBLEM STATEMENT

Chapter 13, Page 500

Step 1

Given:

$$n = \text{Sample size} = 12$$
$$r = \text{Correlation coefficient} = 0.47$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 12 - 2 = 10$ and in the column with $\alpha = 0.01$.

$$t = 2.764$$

The rejection region then contains all values greater than 2.764 and thus we reject H_0 when the t-value is larger than 2.764.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.47\sqrt{12-2}}{\sqrt{1-0.47^2}} \approx 1.684$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 1.684 < 2.764 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a positive association between the variables.

Result

No

PROBLEM STATEMENT

Chapter 13, Page 500

Step 1

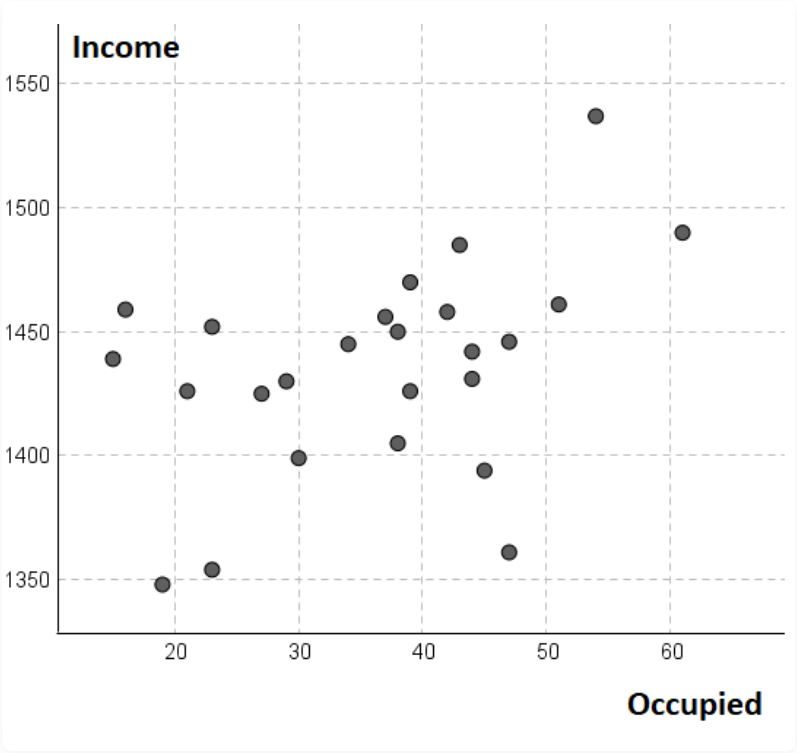
Given:

$$n = \text{Sample size} = 25$$
$$\alpha = \text{Significance level} = 0.10$$

(a)

Scatter diagram

Occupied is on the horizontal axis and Income is on the vertical axis.



Step 2

The pattern in the scatterplot slopes upwards, which indicates that the breakfast revenue seems to increase as the number of occupied rooms increases.

Step 3

(b) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{23 + 47 + 21 + ... + 39}{25} = 36.24$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{1452 + 1361 + 1426 + ... + 1426}{25} = 1435.56$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(23 - 36.24)^2 + ... + (39 - 36.24)^2}{25 - 1}} \approx 12.2450$$
$$s_y = \sqrt{\frac{(1452 - 1435.56)^2 + ... + (1426 - 1435.56)^2}{25 - 1}} \approx 42.7951$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
23	1452	-13.24	16.44	-217.6656
47	1361	10.76	-74.56	-802.2656
21	1426	-15.24	-9.56	145.6944
39	1470	2.76	34.44	95.0544
37	1456	0.76	20.44	15.5344
...	...	...	...	...
39	1426	2.76	-9.56	-26.3856
			Total	**5319.64**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{5319.64}{(25-1)(12.2450)(42.7951)} \\ &\approx 0.4230 \end{aligned}$$

Step 5

(c) Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$\begin{aligned} H_0 &: \rho \leq 0 \text{ (There is no linear correlation)} \\ H_a &: \rho > 0 \text{ (There is a positive linear correlation)} \end{aligned}$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.10$ contains all t-value above the t-value t_0 that has a probability of 0.10 to its right.

$$P(t > t_0) = 0.10$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 25 - 2 = 23$ and in the column with $\alpha = 0.10$.

$$t = 1.319$$

The rejection region then contains all values greater than 1.319 and thus we reject H_0 when the t-value is larger than 1.319.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.4230\sqrt{25-2}}{\sqrt{1-0.4230^2}} \approx 2.2388$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 2.2388 > 1.319 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a positive relationship between revenue and occupied rooms.

Step 7

(d)

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = 0.4230^2 \approx 0.1789 = 17.89\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

17.89% of the variation in the revenue in the restaurant is explained by the variation in the number of rooms occupied.

🎯 **Result**

(a) Yes

(b) $r = 0.4230$

(c) Yes

(d) 17.89%

PROBLEM STATEMENT

Chapter 13, Page 500

Step 1

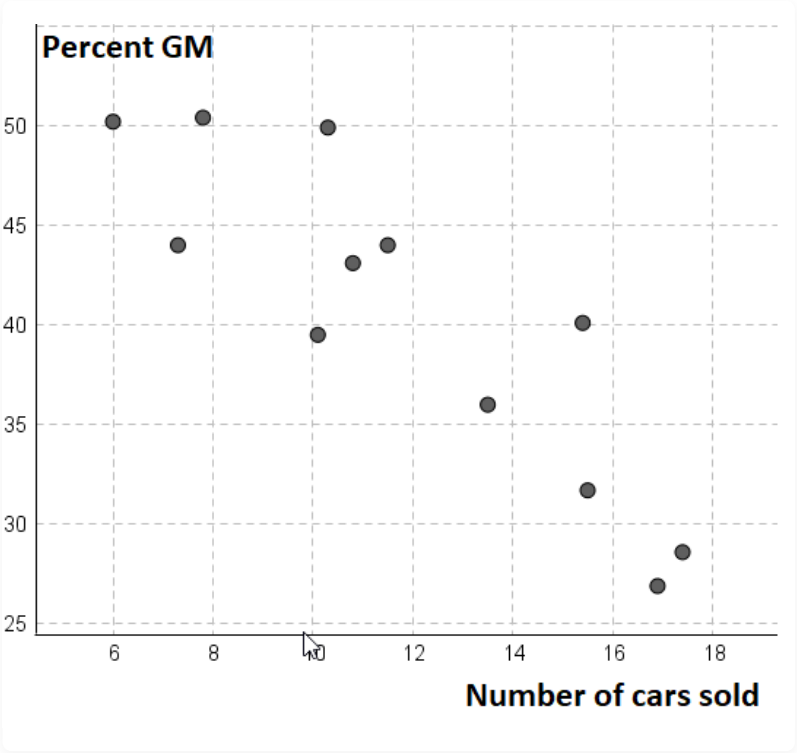
Given:

$$n = \text{Sample size} = 12$$
$$\alpha = \text{Significance level} = 0.01$$

(a)

Scatter diagram

Number of cars sold is on the horizontal axis and Percent GM is on the vertical axis.



Step 2

The pattern in the scatterplot slopes downwards, which indicates that the number of cars sold *indirectly* relates to GM's percent of the market.

Step 3

(b) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{6 + 7.8 + 7.3 + ... + 16.9}{12} = 11.875$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{50.2 + 50.4 + 44 + ... + 26.9}{12} = 40.3667$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(6 - 11.875)^2 + ... + (16.9 - 11.875)^2}{12 - 1}} \approx 3.8490$$
$$s_y = \sqrt{\frac{(50.2 - 40.3667)^2 + ... + (26.9 - 40.3667)^2}{12 - 1}} \approx 42.7951$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
6	50.2	-5.875	9.8333	-57.7706
7.8	50.4	-4.075	10.0333	-40.8857
7.3	44	-4.575	3.6333	-16.6223
10.3	49.9	-1.575	9.5333	-15.0149
10.1	39.5	-1.775	-0.8667	1.5384
10.8	43.1	-1.075	2.7333	-2.9383
11.5	44	-0.375	3.6333	-1.3625
15.4	40.1	3.525	-0.2667	-0.9401
13.5	36	1.625	-4.3667	-7.0959
15.5	31.7	3.625	-8.6667	-31.4168
17.4	28.6	5.525	-11.7667	-65.0110
16.9	26.9	5.025	-13.4667	-67.6702
			Total	**-305.19**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-305.19}{(12-1)(3.8490)(42.7951)} \\ &\approx -0.8807 \end{aligned}$$

Step 5

(c) Given claim: Negative association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \geq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho < 0 \text{ (There is a negative linear correlation)}$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all t-value below the t-value $-t_0$ that has a probability of 0.01 to its left.

$$P(t < -t_0) = 0.01$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 12 - 2 = 10$ and in the column with $\alpha = 0.01$ (which is the negation of the critical value in the table).

$$t = -2.764$$

The rejection region then contains all values smaller than -2.764 and thus we reject H_0 when the t-value is smaller than -2.764.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.8807\sqrt{12-2}}{\sqrt{1-(-0.8807)^2}} \approx -5.8796$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -5.8796 < -2.764 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a negative association between the variables.

Step 7

(d)

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = (-0.8807)^2 \approx 0.7756 = 77.56\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

77.56% of the variation in the GM's market share is explained by the variation in the number of cars sold.

Result

(a) Indirectly

(b) $r = -0.8807$

(c) Yes

(d) 77.56%

PROBLEM STATEMENT

Chapter 13, Page 501

Step 1

Given:

$$n = \text{Sample size} = 32$$
$$r = \text{Correlation coefficient} = -0.363$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Negative association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \geq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho < 0 \text{ (There is a negative linear correlation)}$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 32 - 2 = 30$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.697$$

The rejection region then contains all values smaller than -1.697 and thus we reject H_0 when the t-value is smaller than -1.697.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.363\sqrt{32-2}}{\sqrt{1-(-0.363)^2}} \approx -2.134$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -2.134 < -1.697 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a negative association in the population between the two variables.

Result

Yes

PROBLEM STATEMENT

Chapter 13, Page 501

Step 1

Given:

$$n = \text{Sample size} = 10$$
$$\alpha = \text{Significance level} = 0.05$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{3 + 6 + 5 + ... + 3}{10} = 4.5$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{99 + 104 + 151 + ... + 91}{10} = 111.1$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(3 - 4.5)^2 + ... + (3 - 4.5)^2}{10 - 1}} \approx 1.2693$$
$$s_y = \sqrt{\frac{(99 - 111.1)^2 + ... + (91 - 111.1)^2}{10 - 1}} \approx 24.2920$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X-\overline{X}$	$Y-\overline{Y}$	$(X-\overline{X})(Y-\overline{Y})$
3	99	-1.5	-12.1	18.15
6	104	1.5	-7.1	-10.65
5	151	0.5	39.9	19.95
6	129	1.5	17.9	26.85
6	142	1.5	30.9	46.35
3	111	-1.5	-0.1	0.15
4	74	-0.5	-37.1	18.55
4	91	-0.5	-20.1	10.05
5	119	0.5	7.9	3.95
3	91	-1.5	-20.1	30.15
			Total	163.5

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{163.5}{(10-1)(1.2693)(24.2920)} \\ &\approx 0.5892 \end{aligned}$$

Step 3

(b) **Coefficient of determination**

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = 0.5892^2 \approx 0.3472 = 34.72\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

34.72% of the variation in the amount spent on recreation is explained by the variation in the family size

Step 4

(c) Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$\begin{aligned} H_0 &: \rho \leq 0 \text{ (There is no linear correlation)} \\ H_a &: \rho > 0 \text{ (There is a positive linear correlation)} \end{aligned}$$

Step 5

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 10 - 2 = 8$ and in the column with $\alpha = 0.05$.

$$t = 1.860$$

The rejection region then contains all values greater than 1.860 and thus we reject H_0 when the t-value is larger than 1.860.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.5892\sqrt{10-2}}{\sqrt{1-0.5892^2}} \approx 2.0625$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 2.0625 > 1.860 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a positive association between the amount spend on recreation and family size.

Result

(a) $r = 0.5892$

(b) $r^2 = 34.72\%$

(c) Yes

PROBLEM STATEMENT

Chapter 13, Page 501

Step 1

Given:

$$n = \text{Sample size} = 12$$
$$\alpha = \text{Significance level} = 0.05$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{1.4 + 1.3 + 1.2 + ... + 1.1}{12} = 1.15$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{100 + 110 + 105 + ... + 95}{12} = 96.6667$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(1.4 - 1.15)^2 + ... + (1.1 - 1.15)^2}{12 - 1}} \approx 0.1883$$
$$s_y = \sqrt{\frac{(100 - 96.6667)^2 + ... + (95 - 96.6667)^2}{12 - 1}} \approx 15.7153$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X-\overline{X}$	$Y-\overline{Y}$	$(X-\overline{X})(Y-\overline{Y})$
1.4	100	0.25	3.3333	0.8333
1.3	110	0.15	13.3333	2.0000
1.2	105	0.05	8.3333	0.4167
1.1	120	-0.05	23.3333	-1.1667
1.4	80	0.25	-16.6667	-4.1667
1.0	105	-0.15	8.3333	-1.2500
1.3	110	0.15	13.3333	2.0000
0.8	85	-0.35	-11.6667	4.0833
1.2	15	0.05	8.3333	0.4167
0.9	75	-0.25	-21.6667	5.4167
1.1	70	-0.05	-26.6667	1.3333
1.1	95	-0.05	-1.6667	0.0833
			Total	10

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{10}{(12-1)(0.1883)(15.7153)} \\ &\approx 0.3072 \end{aligned}$$

Step 3

(b) **Coefficient of determination**

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = 0.3072^2 \approx 0.0944 = 9.44\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

9.44% of the variation in the selling price is explained by the variation in the home size.

Step 4

(c) Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$\begin{aligned} H_0 &: \rho \leq 0 \text{ (There is no linear correlation)} \\ H_a &: \rho > 0 \text{ (There is a positive linear correlation)} \end{aligned}$$

Step 5

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 12 - 2 = 10$ and in the column with $\alpha = 0.05$.

$$t = 1.812$$

The rejection region then contains all values greater than 1.812 and thus we reject H_0 when the t-value is larger than 1.812.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.3072\sqrt{12-2}}{\sqrt{1-0.3072^2}} \approx 1.0208$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 1.0208 < 1.812 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a positive association between the size of the home and the selling price.

Result

(a) $r = 0.3072$

(b) $r^2 = 9.44\%$

(c) No

PROBLEM STATEMENT

Chapter 13, Page 501

Step 1

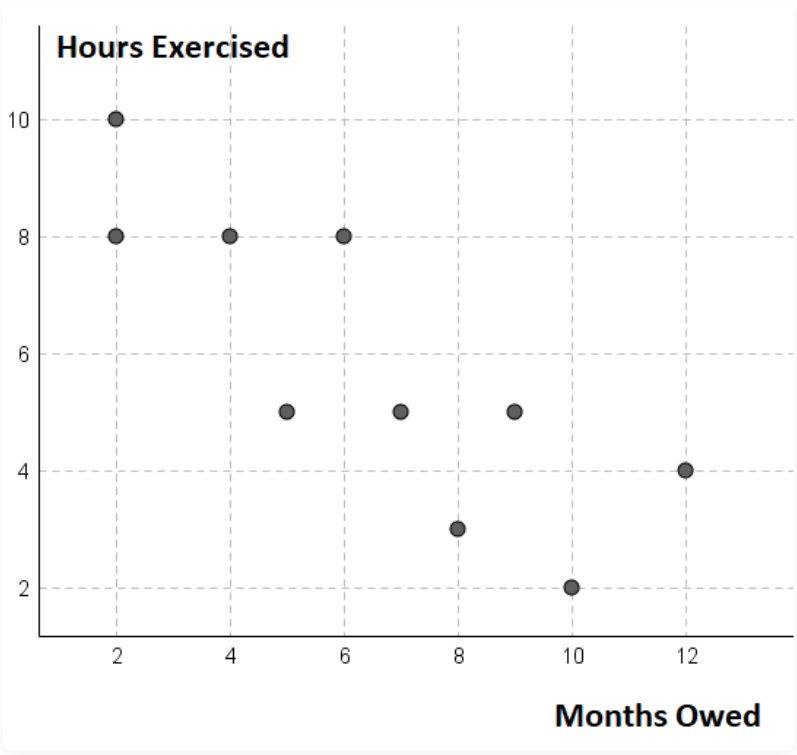
Given:

$$n = \text{Sample size} = 10$$
$$\alpha = \text{Significance level} = 0.01$$

(a)

Scatter diagram

Months owed is on the horizontal axis and hours exercised is on the vertical axis.



Step 2

The pattern in the scatter diagram slopes downwards, which indicates that there is a negative association between months owed and hours exercised.

Step 3

(b)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{12 + 2 + 6 + ... + 5}{10} = 6.5$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{4 + 10 + 8 + ... + 5}{10} = 5.8$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(12 - 6.5)^2 + ... + (5 - 6.5)^2}{10 - 1}} \approx 3.3417$$
$$s_y = \sqrt{\frac{(4 - 5.8)^2 + ... + (5 - 5.8)^2}{10 - 1}} \approx 2.5734$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
12	4	5.5	-1.8	-9.9
2	10	-4.5	4.2	-18.9
6	8	-0.5	2.2	-1.1
9	5	2.5	-0.8	-2
7	5	0.5	-0.8	-0.4
2	8	-4.5	2.2	-9.9
8	3	1.5	-2.8	-4.2
4	8	-2.5	2.2	-5.5
10	2	3.5	-3.8	-13.3
5	5	-1.5	-0.8	1.2
			Total	**-64**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-64}{(10-1)(3.3417)(2.5734)} \\ &\approx -0.8269 \end{aligned}$$

Step 5

(c) Given claim: Negative association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \geq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho < 0 \text{ (There is a positive linear correlation)}$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all t-value below the t-value $-t_0$ that has a probability of 0.01 to its left.

$$P(t < -t_0) = 0.01$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 10 - 2 = 8$ and in the column with $\alpha = 0.01$ (which is the negation of the critical value in the table).

$$t = -2.896$$

The rejection region then contains all values smaller than -2.896 and thus we reject H_0 when the t-value is smaller than -2.896.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.8269\sqrt{10-2}}{\sqrt{1-(-0.8269)^2}} \approx -4.1590$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -4.1590 < -2.896 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a negative association between the variables.

🎯 **Result**

(a) Negative association

(b) $r = -0.8269$

(c) Yes

PROBLEM STATEMENT

Chapter 13, Page 502

Step 1

Given:

$$\begin{aligned} n &= \text{Sample size} = 20 \\ SSE &= 100 \\ SS\ \text{total} &= 400 \\ \hat{y} &= 15 - 5x \end{aligned}$$

(a) The standard error of the estimate is the square root of the error sum of squares divided by the sample size decreased by 2.

$$s = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{100}{20-2}} \approx 2.3570$$

Step 2

(b) Since the equality $SS\ \text{total} = SSR + SSE$ holds:

$$SSR = SS\ \text{total} - SSE = 400 - 100 = 300$$

The coefficient of determination is the ratio of the regression sum of squares and the total sum of squares:

$$r^2 = \frac{SSR}{SS\ \text{total}} = \frac{300}{400} = \frac{3}{4} = 0.75 = 75\%$$

Step 3

(c) The slope is the coefficient of x in the regression equation:

$$b = -5$$

Since the slope b is negative, the correlation coefficient needs to be negative as well.

The correlation coefficient is then the negative square root of the coefficient of determination:

$$r = -\sqrt{r^2} = -\sqrt{0.75} \approx -0.8660$$

🎯 **Result**

(a) 2.3570

(b) 75%

(c) -0.8660

PROBLEM STATEMENT

Chapter 13, Page 502

Step 1

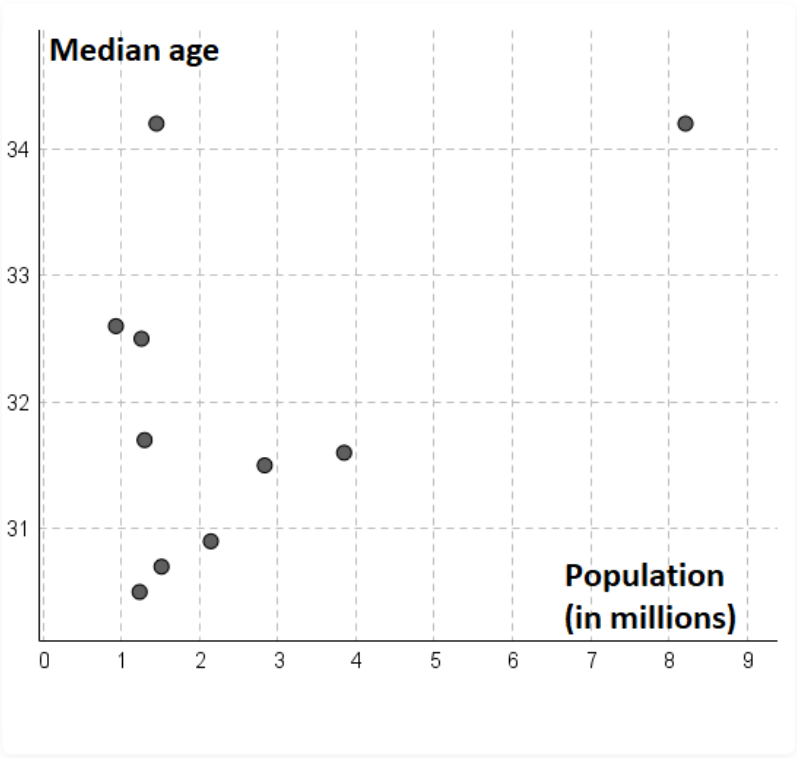
Given:

$$n = \text{Sample size} = 10$$
$$\alpha = \text{Significance level} = 0.10$$
$$\hat{y} = 31.4 + 0.272x$$

(a)

Scatter diagram

Population is on the horizontal axis and Median age is on the vertical axis.



Step 2

(b) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2.833 + 1.233 + 2.144 + ... + 0.930}{10} = 2.4718$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{31.5 + 30.5 + 30.9 + ... + 32.6}{10} = 32.04$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(2.833 - 2.4718)^2 + ... + (0.930 - 2.4718)^2}{10 - 1}} \approx 2.2071$$
$$s_y = \sqrt{\frac{(31.5 - 32.04)^2 + ... + (32.6 - 32.04)^2}{10 - 1}} \approx 1.3302$$

Step 3

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
2.833	31.5	0.3612	-0.54	-0.1950
1.233	30.5	-1.2388	-1.54	1.9078
2.144	30.9	-0.3278	-1.14	0.3737
3.849	31.6	1.3772	-0.44	-0.6060
8.214	34.2	5.7422	2.16	12.4032
1.448	34.2	-1.0238	2.16	-2.2114
1.513	30.7	-0.9588	-1.34	1.2848
1.297	31.7	-1.1748	-0.34	0.3994
1.257	32.5	-1.2148	0.46	-0.5588
0.930	32.6	-1.5418	0.56	-0.8634
			Total	**11.9342**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{11.9342}{(10-1)(2.2071)(1.3302)} \\ &\approx 0.4517 \end{aligned}$$

Step 4

(c) The slope of the regression line is the coefficient of x (population):

$$b = 0.272$$

The slope represents the average increase (or decreased when negative) of the y-variable per unit of x.

The median age increases on average by 0.272 years per one million of population.

Step 5

(d) Evaluate the regression equation at $x = 2.5$:

$$\hat{y} = 31.4 + 0.272(2.5) = 32.08$$

Thus the estimate median age is 32.08 years.

Step 6

(e) The P-values are given in the column "P" of the regression output.

Since $P = 0.000$ is small for the constant, the constant appears to be nonzero.

Since $P = 0.190$ is large for the slope (greater than 0.05), the slope appears to be approximately zero.

Step 7

(f) Given claim: Slope is zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

The value of the slope is given in the row "Population" and in the column "P" of the regression output:

$$P = 0.190$$

If the P-value is smaller than the significance level α, then we reject the null hypothesis.

$$P = 0.190 > 0.10 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a relationship between the variables.

🎯 **Result**

(a) Scatter diagram

(b) $r = 0.4517$

(c) The median age increases on average by 0.272 years per one million of population.

(d) 32.08 years

(e) Constant is nonzero, Slope is approximately zero.

(f) No

PROBLEM STATEMENT

Chapter 13, Page 502

Step 1

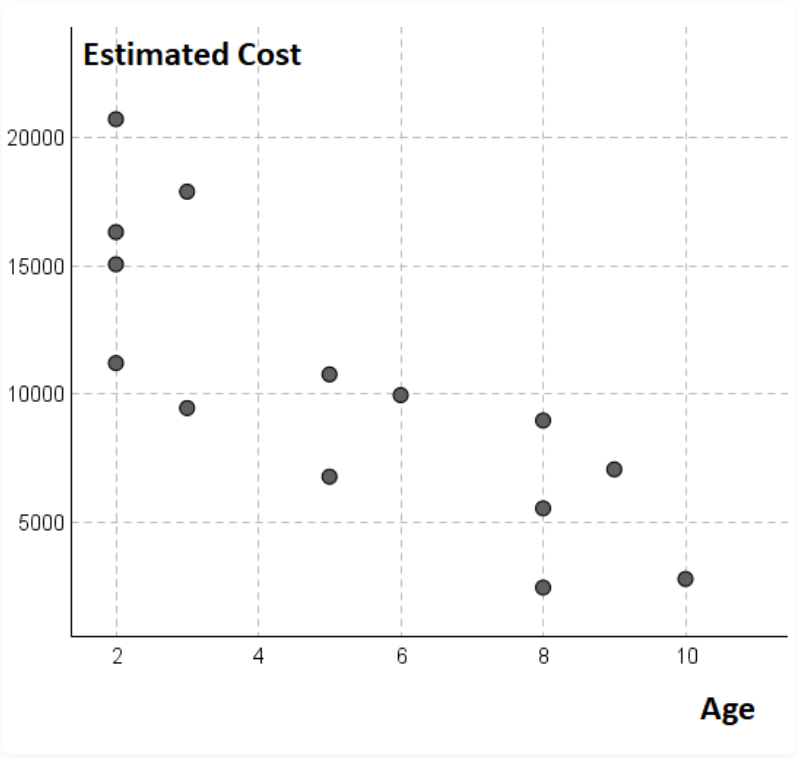
Given:

$$n = \text{Sample size} = 14$$
$$\alpha = \text{Significance level} = 0.10$$
$$\hat{y} = 18358 - 1534x$$

(a)

Scatter diagram

Age is on the horizontal axis and Estimated Cost is on the vertical axis.



Step 2

(b) Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{8 + 3 + 6 + ... + 2}{14} = 5.2143$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{5555 + 17888 + ... + 20705}{14} = 10360.9286$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(8 - 5.2143)^2 + ... + (2 - 5.2143)^2}{14 - 1}} \approx 2.9399$$
$$s_y = \sqrt{\frac{(5555 - 10360.9286)^2 + ... + (20705 - 10360.9286)^2}{14 - 1}} \approx 5482.3373$$

Step 3

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
8	5555	2.7857	-4805.9286	-13387.8753
3	178888	-2.2143	7527.0714	-16667.1942
6	9963	0.7857	-397.9286	-312.6525
5	6793	-0.2143	-3567.9286	764.6071
5	10774	-0.2143	413.0714	-88.5212
2	16310	-3.2143	5949.0714	-19122.1002
8	2475	2.7857	-7885.9286	-21967.8313
10	2808	4.7857	-7552.9286	-36146.0504
9	7030	3.7857	-3287.9286	-12447.1113
8	8978	2.7857	-1382.9286	-3852.4242
2	11213	-3.2143	852.0714	-2738.8131
3	9463	-2.2143	-897.9286	1988.2833
2	15055	-3.2143	4694.0714	-15088.1537
2	20705	-3.2143	10344.0714	-33248.9487
			Total	**-172314.7857**

Finally, we evaluate the formula of the correlation coefficient:

$$
\begin{aligned}
r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\
&= \frac{-172314.7857}{(14-1)(2.9399)(5482.3373)} \\
&\approx -0.8224
\end{aligned}
$$

Step 4

(c) The slope of the regression line is the coefficient of x (age):

$$b = -1534$$

The slope represents the average increase (or decreased when negative) of the y-variable per unit of x.

The estimated cost decreases on average by $\$1534$ per year.

Step 5

(d) Evaluate the regression equation at $x = 5$:

$$\hat{y} = 18358 - 1534(5) = 10688$$

Thus the estimated cost of a five-year-old car is $\$10,688$.

Step 6

(e) The P-values are given in the column "P" of the regression output.

Since $P = 0.000$ is small for the constant, the constant appears to be nonzero.

Since $P = 000$ is small for the slope, the slope appears to be nonzero as well.

Step 7

(f) Given claim: Slope is zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

The value of the slope is given in the row "Age" and in the column "P" of the regression output:

$$P = 0.000$$

If the P-value is smaller than the significance level α, then we reject the null hypothesis.

$$P = 0.000 < 0.10 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a relationship between the variables.

Result

(a) Scatter diagram

(b) $r = -0.8224$

(c) The estimated cost decreases on average by $\$1534$ per year.

(d) $\$10,688$

(e) Constant and slope are nonzero

(f) Yes

PROBLEM STATEMENT

Chapter 13, Page 503

Step 1

Given:

$$\begin{aligned} n &= \text{Sample size} = 15 \\ c &= \text{Confidence interval} = 95\% \\ x_0 &= 7 \end{aligned}$$

(a)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x} &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{9 + 9 + 3 + ... + 6}{15} = 7.1333 \\ \overline{y} &= \frac{\sum_{i=1}^{n} y_i}{n} = \frac{5.1 + 8.0 + 9.7 + ... + 7.8}{15} = 7.9067 \end{aligned}$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$\begin{aligned} s_x &= \sqrt{\frac{(9 - 7.1333)^2 + ... + (6 - 7.1333)^2}{15 - 1}} \approx 2.2949 \\ s_y &= \sqrt{\frac{(5.1 - 7.9067)^2 + ... + (7.8 - 7.9067)^2}{15 - 1}} \approx 1.5163 \end{aligned}$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$	$(X - \overline{X})^2$
9	5.1	1.8667	-2.8067	-5.2393	3.4846
9	8	1.8667	0.0933	0.1742	3.4846
3	9.7	-4.1333	1.7933	7.4122	17.0842
10	7.8	2.8667	-0.1067	-0.3059	8.2180
5	77	-2.1333	-0.2067	0.4410	4.5510
10	5.5	2.8667	-2.4067	-6.8993	8.2180
7	8.3	-0.1333	0.3933	-0.0524	0.0178
11	5.5	3.8667	-2.4067	-9.3060	14.9514
6	10.3	-1.1333	2.3933	-2.7123	1.2844
6	8	-1.1333	0.0933	-0.1057	1.2844
4	8.8	-3.1333	0.8933	-2.7990	9.8176
7	9.4	-0.1333	1.4933	-0.1991	0.0178
7	8.6	-0.1333	0.6933	-0.0924	0.0178
7	8.1	-0.1333	0.1933	-0.0258	0.0178
6	7.8	-1.1333	-0.1067	0.1209	1.2844
			Total	**-34.4133**	**73.7333**

Finally, we evaluate the formula of the correlation coefficient:

$$
\begin{aligned}
r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\
&= \frac{-34.4133}{(15-1)(2.2949)(1.5163)} \\
&\approx -0.7064
\end{aligned}
$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.7064\frac{1.5163}{2.2949} = -0.4667$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 7.9067 - (-0.4667) \cdot 7.1333 = 11.2360$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 11.2360 - 0.4667x$$

The slope $b = -0.4667$ is negative, which implies that the amount of the winning bid tends to decrease as there are more bidders.

Step 4

(b) Evaluate the regression equation at $x = x_0 = 7$:

$$\hat{y} = a + bx = 11.2360 - 0.4667(7) = 7.9689$$

Thus the estimated amount of the winning bid is 7.9689 million dollars.

Step 5

(c)

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y - \hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y - \hat{Y}$	$(Y - \hat{Y})^2$
9	5.1	7.0357	-1.9357	3.7469
9	8	7.0357	0.9643	0.9299
3	9.7	9.8359	-0.1359	0.0185
10	7.8	6.5690	1.2310	1.5154
5	77	8.9025	-1.2025	1.4460
10	5.5	6.5690	-1.0690	1.1428
7	8.3	7.9691	0.3309	0.1095
11	5.5	6.1023	-0.6023	0.3628
6	10.3	8.4358	1.8642	3.4752
6	8	8.4358	-0.4358	0.1899
4	8.8	9.3692	-0.5692	0.3240
7	9.4	7.9691	1.4309	2.0475
7	8.6	7.9691	0.6309	0.3980
7	8.1	7.9691	0.1309	0.0171
6	7.8	8.4358	-0.6358	0.4042
			Total	**16.1277**

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n - 2$.

$$s = \sqrt{\frac{\sum (Y - \hat{Y})^2}{n - 2}} = \sqrt{\frac{16.1277}{15 - 2}} \approx 1.1138$$

Step 6

Prediction interval for an individual

Evaluate the regression equation at $x = x_0 = 7$:

$$\hat{y} = a + bx = 11.2360 - 0.4667(7) = 7.9689$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 15 - 2 = 13$ and $c = 95\%$:

$$t_c = \pm 2.160$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 2.160(1.1138)\sqrt{1 + \frac{1}{15} + \frac{(7 - 7.1333)^2}{73.7333}} \\ &\approx 2.4850 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 7.9689 - 2.4850 = 5.4839 \\ \hat{y} + E &= 7.9689 + 2.4850 = 10.4539 \end{aligned}$$

Step 7

(d)

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = (-0.7064)^2 \approx 0.4990 = 49.90\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

49.90% of the variation in the value of the winning bid is explained by the variation in the number of bidders.

🎯 **Result**

(a) Decrease

(b) 7.9689 million dollars

(c) $(5.4839, 10.4539)$ million dollars

(d) 49.90% of the variation in the value of the winning bid is explained by the variation in the number of bidders.

PROBLEM STATEMENT

Chapter 13, Page 503

Step 1

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{9.0 + 94.4 + 27.3 + ... + 34.4}{15} = 79.5867$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{10.8 + 11.3 + 11.2 + ... + 10.8}{15} = 11.875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(9.0 - 79.5867)^2 + ... + (34.4 - 79.5867)^2}{15 - 1}} \approx 47.2391$$
$$s_y = \sqrt{\frac{(10.8 - 11.875)^2 + ... + (10.8 - 11.875)^2}{15 - 1}} \approx 0.3058$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
9	10.8	-70.5867	-10.67	7.5316
94.4	11.3	14.8133	0.3933	5.8261
27.3	11.2	-52.2867	0.2933	-15.3357
$\ldots$	$\ldots.$	$\ldots$	$\ldots$	$\ldots$
34.4	10.8	-45.1867	-0.1067	4.8214
			Total	94.2613

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1) s_x s_y} \\ &= \frac{94.2613}{(15-1)(47.2391)(0.3058)} \\ &\approx 0.4661 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r \frac{s_y}{s_x} = 0.4661 \frac{0.3058}{47.2391} = 0.0030$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 10.9067 - 0.0030 \cdot 79.5867 = 10.6665$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 10.6665 + 0.0030x$$

Step 4

(b) Since we did not see a formula for calculating the standard deviation of the slope, I will use a hypothesis test about the correlation coefficient (which will give the same results).

Given claim: Positive slope

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 5

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 15 - 2 = 13$ and in the column with $\alpha = 0.05$.

$$t = 1.771$$

The rejection region then contains all values greater than 1.771 and thus we reject H_0 when the t-value is larger than 1.771.\

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.4661\sqrt{10-2}}{\sqrt{1-0.4661^2}} \approx 1.4901$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 1.4901 < 1.771 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the slope of the regression line is positive.

Step 6

(c) **Coefficient of determination**

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = 0.4661^2 \approx 0.2172 = 21.72\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

21.72% of the variation in the price per share is explained by the variation in the size.

Since the percentage of 21.72% is quite small, Mr. Profit should not be satisfied with using the size of the offering as the independent variable.

🎯 Result

(a) $\hat{y} = 10.6665 + 0.0030x$

(b) There is not sufficient evidence to support the claim that the slope of the regression line is positive.

(c) No.

PROBLEM STATEMENT

Chapter 13, Page 504

Step 1

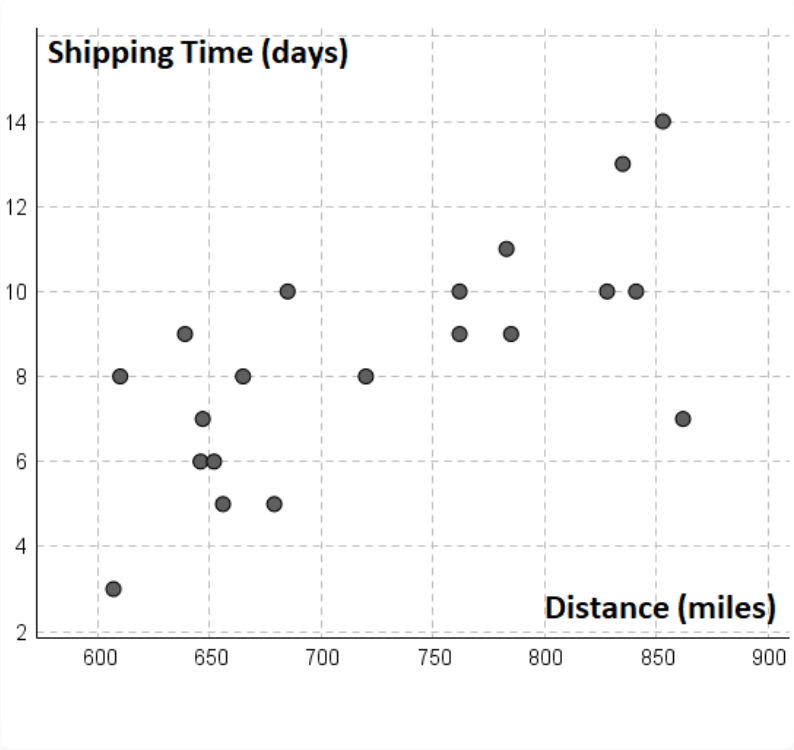
Given:

$$n = \text{Sample size} = 20$$

(a)

Scatter diagram

Distance is on the horizontal axis and shipping time is on the vertical axis.



Step 2

We note that the pattern in the scatter diagram slopes downwards, which indicates that there is a relationship between how many miles a shipment has to go and the time it takes to arrive at its destination.

Step 3

(b)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{656 + 853 + 646 + ... + 828}{20} = 725.85$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{5 + 14 + 6 + ... + 10}{20} = 8.4$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(656 - 725.85)^2 + ... + (828 - 725.85)^2}{20 - 1}} \approx 87.4512$$
$$s_y = \sqrt{\frac{(5 - 8.4)^2 + ... + (10 - 8.4)^2}{20 - 1}} \approx 2.7028$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
656	5	-69.85	-3.4	237.49
853	14	127.15	5.6	712.04
646	6	-79.85	-2.4	191.64
...	...	...	...	...
828	10	102.15	1.6	163.44
			Total	**3108.2**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{3108.2}{(20-1)(87.4512)(2.7028)} \\ &\approx 0.6921 \end{aligned}$$

Step 5

Hypothesis test correlation

Given claim: Positive correlation

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$\begin{aligned} H_0 &: \rho \leq 0 \text{ (There is no linear correlation)} \\ H_a &: \rho > 0 \text{ (There is a positive linear correlation)} \end{aligned}$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 20 - 2 = 18$ and in the column with $\alpha = 0.05$.

$$t = 1.734$$

The rejection region then contains all values greater than 1.734 and thus we reject H_0 when the t-value is larger than 1.734.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.6921\sqrt{20-2}}{\sqrt{1-0.6921^2}} \approx 4.0681$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 4.0681 > 1.734 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a positive correlation between distance and time.

Step 7

(c)

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = 0.6921^2 \approx 0.4790 = 47.90\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

47.90% of the variation in the shipping time is explained by the variation in the distance.

Step 8

(d)

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.6921\frac{2.7028}{87.4512} = 0.0214$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 8.4 - 0.0214 \cdot 725.85 = -7.1264$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -7.1264 + 0.0214x$$

Step 9

Standard error of the estimate

We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y - \hat{y})^2$ for each given y-value.

X	Y	$\hat{Y}$	$Y - \hat{Y}$	$(Y - \hat{Y})^2$
656	5	6.912	-1.912	3.6557
853	14	11.1278	2.8722	8.2495
646	6	6.698	-0.698	0.4872
...	...	...	...	...
828	10	10.5928	-0.5928	0.3514
			Total	**72.3145**

The standard deviation of the residuals is then the square root of the sum of the squared residuals divided by $n - 2$.

$$s = \sqrt{\frac{\sum(Y - \hat{Y})^2}{n - 2}} = \sqrt{\frac{72.3145}{20 - 2}} \approx 3.3784$$

Step 10

(e) We note that the coefficient of determination of 47.90% is not very large and thus predictions are likely be inaccurate.

Moreover, the standard error of the estimate is 2.0044. Since a variation of 2 days in the shipping time is large, we would not recommend using the regression equation to predict shipping time.

Result

(a) Yes

(b) $r = 0.6921$, Yes

(c) 47.90% of the variation in the shipping time is explained by the variation in the distance.

(d) $s = 2.0044$

(e) No

PROBLEM STATEMENT

Chapter 13, Page 504

Step 1

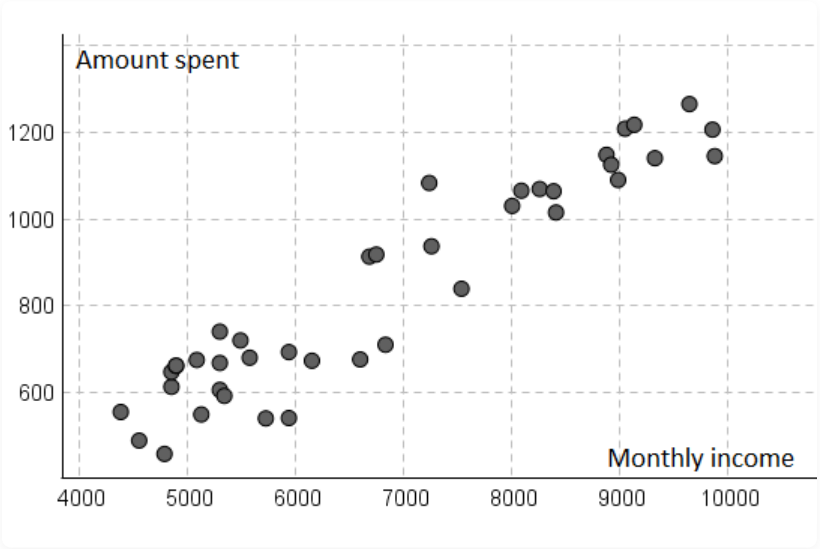
Given:

$$n = \text{Sample size} = 40$$

(a)

Scatter diagram

Amount spent is on the horizontal axis and Monthly income is on the vertical axis.



Step 2

We note that the pattern in the scatter diagram slopes upwards, which indicates that there is a relationship between amount spent and monthly income.

Step 3

(b)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{y} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{555 + 489 + ... + 1206 + 1145}{40} = 840.625$$
$$\overline{x} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{4388 + 4558 + ... + 9862 + 9883}{40} = 6834.675$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_y = \sqrt{\frac{(555 - 840.625)^2 + ... + (1145 - 840.625)^2}{40 - 1}} \approx 247.9076$$
$$s_x = \sqrt{\frac{(4388 - 6834.675)^2 + ... + (9883 - 6834.675)^2}{40 - 1}} \approx 1749.4638$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

Y	X	$Y - \overline{Y}$	$X - \overline{X}$	$(X - \overline{X})(Y - \overline{Y})$
555	4388	-28.625	-2446.675	698831.5469
489	4558	-351.625	-2276.675	800535.8469
...	...	...	...	...
1206	9862	365.375	3027.325	1106108.8719
1145	9883	304.375	3048.325	927833.9219
			Total	**15979888.125**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1) s_x s_y} \\ &= \frac{15979888.125}{(40-1)(247.9076)(1749.4638)} \\ &\approx 0.9447 \end{aligned}$$

Step 5

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r \frac{s_y}{s_x} = 0.9447 \frac{247.9076}{1749.4638} = 0.1339$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 840.625 - 6.6670 \cdot 6834.675 = -74.3665$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -74.3665 + 0.1339x$$

The amount spent increases on average by 0.1339 dollars per dollar of the monthly income.

Step 6

(c)

Hypothesis test correlation

Given claim: Positive correlation

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 7

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 40 - 2 = 38$ and in the column with $\alpha = 0.05$.

$$t = 1.686$$

The rejection region then contains all values greater than 1.686 and thus we reject H_0 when the t-value is larger than 1.686.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9447\sqrt{40-2}}{\sqrt{1-0.9447^2}} \approx 17.7581$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 17.7581 > 1.686 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a correlation coefficient is greater than 0.

🎯 **Result**

(a) Amount spent is on the horizontal axis and Monthly income is on the vertical axis.

(b) $\hat{y} = 0.1339x - 74.3665$

The amount spent increases on average by 0.1339 dollars per dollar of the monthly income.

(c) $r = 0.9447$, Yes

PROBLEM STATEMENT

Chapter 13, Page 505

Step 1

Given:

$$n = \text{Sample size} = 30$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{3.14 + 3.36 + ... + 17.65 + 17.36}{30} = 10.6777$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{20.00 + 22.01 + ... + 77.91 + 80.00}{30} = 52.5190$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(3.14 - 10.6777)^2 + ... + (17.36 - 10.6777)^2}{30 - 1}} \approx 5.4838$$
$$s_y = \sqrt{\frac{(20.00 - 52.5190)^2 + ... + (80.00 - 52.5190)^2}{30 - 1}} \approx 16.2764$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
3.14	20	-7.5377	-32.5190	245.1185
3.36	22.01	-7.3177	-30.5090	223.2557
...	...	...	...	..
17.65	77.91	6.9723	25.391	177.0337
80	17.36	27.481	6.6823	183.6363
			Total	2100.26

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{2100.26}{(30-1)(5.4838)(16.2764)} \\ &\approx 0.8114 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 9.68276\frac{16.2764}{5.4838} = 2.408$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 52.5190 - 2.408 \cdot 10.6777 = 26.81$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 26.81 + 2.408x$$

Step 4

(b) A hypothesis test for the slope is equivalent with a hypothesis test for the correlation.

Hypothesis test correlation

Given claim: No significant relationship

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho = 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho \neq 0 \text{ (There is a linear correlation)}$$

Step 5

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 30 - 2 = 28$ and in the column with $\alpha/2 = 0.05/2 = 0.025$.

$$t = 2.048$$

The rejection region then contains all values greater than 2.048 and thus we reject H_0 when the t-value is larger than 2.048.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.8114\sqrt{30-2}}{\sqrt{1-0.8114^2}} \approx 7.346$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 7.346 > 2.048 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the slope is significant.

Step 6

(c) **Coefficient of determination**

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = 0.8114^2 \approx 0.6583 = 65.83\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

65.83% of the variation in the price per share is explained by the variation in the dividend.

Step 7

(d) **Hypothesis test correlation**

Given claim: Correlation is greater than 0

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 8

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 30 - 2 = 28$ and in the column with $\alpha = 0.05$.

$$t = 1.701$$

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.8114\sqrt{30-2}}{\sqrt{1-0.8114^2}} \approx 7.346$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 7.346 > 1.701 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the correlation coefficient is greater than 0.

🎯 **Result**

(a) $\hat{y} = 26.81 + 2.408x$

(b) Significant

(c) 65.83%

(d) Yes

PROBLEM STATEMENT

Chapter 13, Page 505

Step 1

Given:

$$\hat{y} = 12.7 + 0.000114x$$
$$\alpha = \text{Significance level} = 0.05$$

(a) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table in the given output:

$$d.f._{total} = 19$$

The total degrees of freedom is the total sample size decreased by 1, which implies that the total sample size is the total degrees of freedom increased by 1.

$$n = d.f._{total} + 1 = 19 + 1 = 20$$

Thus there are 20 states in the sample.

Step 2

(b) The error sum of squares is given in the row "Residual Error" and in the column "SS" of the ANOVA table in the given output:

$$SSE = 12,054$$

The standard error of the estimate is the square root of the error sum of squares divided by the sample size decreased by 2.

$$s = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{12,054}{20-2}} \approx 25.8779$$

Step 3

(c) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table in the given output:

$$SSR = 10,354$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table in the given output:

$$SS\ total = 22,408$$

The coefficient of determination is the ratio of the regression sum of squares to the total sum of squares.

$$r^2 = \frac{SSR}{SS\ total} = \frac{10,354}{22,408} \approx 0.4621 = 46.21\%$$

Step 4

(d) The slope of the regression line is the coefficient of x (unemployed) in the regression equation:

$$b = 0.000114$$

Since the slope $b = 0.000114$ is positive, the correlation coefficient needs to be positive as well.

The correlation coefficient is then the positive square root of the coefficient of determination:

$$r = +\sqrt{r^2} = +\sqrt{0.4621} \approx 0.6798$$

Step 5

(e) Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 20 - 2 = 18$ and in the column with $\alpha = 0.05$.

$$t = 1.734$$

The rejection region then contains all values greater than 1.734 and thus we reject H_0 when the t-value is larger than 1.734.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.6798\sqrt{20-2}}{\sqrt{1-0.6798^2}} \approx 3.933$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 3.933 > 1.734 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a positive association between fatalities and the number unemployed.

Result

(a) 20 states

(b) $s = 25.8779$

(c) $r^2 = 46.21\%$

(d) $r = 0.6798$

(e) Yes

PROBLEM STATEMENT

Chapter 13, Page 505

Step 1

Given:

$$\hat{y} = -37,186 + 65.0x$$
$$\alpha = \text{Significance level} = 0.05$$

(a) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table in the given output:

$$d.f._{total} = 34$$

The total degrees of freedom is the total sample size decreased by 1, which implies that the total sample size is the total degrees of freedom increased by 1.

$$n = d.f._{total} + 1 = 34 + 1 = 35$$

Thus there are 35 homes in the sample.

Step 2

(b) The error sum of squares is given in the row "Residual Error" and in the column "SS" of the ANOVA table in the given output:

$$SSE = 982,687,392$$

The standard error of the estimate is the square root of the error sum of squares divided by the sample size decreased by 2.

$$s = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{982,687,392}{35-2}} \approx 5456.9594$$

Step 3

(c) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table in the given output:

$$SSR = 13,548,662,082$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table in the given output:

$$SS\ total = 14,531,349,474$$

The coefficient of determination is the ratio of the regression sum of squares to the total sum of squares.

$$r^2 = \frac{SSR}{SS\ total} = \frac{13,548,662,082}{14,531,349,474} \approx 0.9324 = 93.24\%$$

Step 4

(d) The slope of the regression line is the coefficient of x (unemployed) in the regression equation:

$$b = 65.0$$

Since the slope $b = 65.0$ is positive, the correlation coefficient needs to be positive as well.

The correlation coefficient is then the positive square root of the coefficient of determination:

$$r = +\sqrt{r^2} = +\sqrt{0.9324} \approx 0.9656$$

Step 5

(e) Given claim: Positive association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 35 - 2 = 33$ and in the column with $\alpha = 0.05$.

$$t = 1.692$$

The rejection region then contains all values greater than 1.692 and thus we reject H_0 when the t-value is larger than 1.692.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9656\sqrt{35-2}}{\sqrt{1-0.9656^2}} \approx 21.332$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 21.332 > 1.692 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a positive association between the market value of homes and the size of the home in square feet.

Result

(a) 35 homes

(b) $s = 5456.9594$

(c) $r^2 = 93.24\%$

(d) $r = 0.9656$

(e) Yes

PROBLEM STATEMENT

Chapter 13, Page 506

Step 1

Given:

$$n = \text{Sample size} = 8$$
$$\alpha = \text{Significance level} = 0.05$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{23.1 + 13.2 + 24.2 + ... + 20.1}{8} = 17.4875$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{8 + 15.6 + 31.2 + ... + 3.2}{8} = 13.825$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(23.1 - 17.4875)^2 + ... + (20.1 - 17.4875)^2}{8-1}} \approx 6.9503$$
$$s_y = \sqrt{\frac{(8 - 13.825)^2 + ... + (3.2 - 13.825)^2}{8-1}} \approx 12.7228$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X-\overline{X}$	$Y-\overline{Y}$	$(X-\overline{X})(Y-\overline{Y})$
23.1	8	5.6125	-5.825	-32.6928
13.2	15.6	-4.2875	1.775	-7.6103
24.2	31.2	6.712	17.375	116.6297
11.1	2.5	-6.3875	-11.325	72.3384
10.1	35.4	-7.3875	21.575	-159.3853
10.8	6	-6.6875	-7.825	52.3297
27.3	8.7	9.8125	-5.125	-50.2891
20.1	3.2	2.6125	-10.625	-27.7578
			Total	-36.4375

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-36.4375}{(8-1)(6.9503)(12.7228)} \\ &\approx -0.0589 \end{aligned}$$

Step 3

Given claim: Correlation is greater than 0

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 4

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 8 - 2 = 6$ and in the column with $\alpha = 0.05$.

$$t = 1.943$$

The rejection region then contains all values greater than 1.943 and thus we reject H_0 when the t-value is larger than 1.943.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.0589\sqrt{8-2}}{\sqrt{1-(-0.0589)^2}} \approx -0.1445$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -0.1445 < 1.943 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the population correlation is greater than zero.

Step 5

(b) **Regression equation**

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = -0.0589\frac{12.7228}{6.9503} = -0.1078$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 13.825 - (-0.1078) \cdot 17.4875 = 15.7094$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 15.7094 - 0.1078x$$

Step 6

We have not been given a formula on how to calculate the standard deviation of the slope and thus we cannot conduct a hypothesis test for the slope. We will execute a hypothesis test for the correlation instead, which will have the same results as a hypothesis test for the slope.

Given claim: Slope is negative

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \geq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho < 0 \text{ (There is a negative linear correlation)}$$

Step 7

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value t_0 that has a probability of 0.05 to its left.

$$P(t < t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 8 - 2 = 6$ and in the column with $\alpha = 0.05$.

$$t = -1.943$$

The rejection region then contains all values less than -1.943 and thus we reject H_0 when the t-value is less than -1.943.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.0589\sqrt{8-2}}{\sqrt{1-(-0.0589)^2}} \approx -0.1445$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -0.1445 > -1.943 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the slope of the regression line is negative.

Step 8

(c) We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y - \hat{y})$ for each given y-value, which is the residual.

X	Y	$\hat{Y}$	$Y - \hat{Y}$
23.1	8	13.2192	-5.2192
13.2	15.6	14.2864	1.3136
24.2	31.2	13.1006	18.0994
11.1	2.5	14.5128	-12.0128
10.1	35.4	14.6206	20.7794
10.8	6	14.5452	-8.5452
27.3	8.7	12.7665	-4.0665
20.1	3.2	13.5426	-10.3426

We note that 20.7794 is the largest residual, which belongs to the company L-3 Communications.

Result

(a) $r = -0.0589$, There is not sufficient evidence to support the claim that the population correlation is greater than zero.

(b) $\hat{y} = 15.7094 - 0.1078x$, No

(c) L-3 Communications

PROBLEM STATEMENT

Chapter 13, Page 506

Step 1

Given:

$$n = \text{Sample size} = 12$$
$$\alpha = \text{Significance level} = 0.05$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2 + 1.6 + 1.6 + ... + 1.4}{12} = 1.65$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{2017 + 922 + 1064 + ... + 869}{12} = 1550$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(2 - 1.65)^2 + ... + (1.4 - 1.65)^2}{12 - 1}} \approx 0.3317$$
$$s_y = \sqrt{\frac{(2017 - 1550)^2 + ... + (869 - 1550)^2}{12 - 1}} \approx 559.5009$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
2	2017	0.35	467	163.45
1.6	922	-0.05	-628	31.4
1.6	1064	-0.05	-486	24.3
1.8	1942	0.15	392	58.8
2	2137	0.35	587	205.45
1.2	1012	-0.45	-538	242.1
2	2197	0.35	647	226.45
1.6	1387	-0.05	-163	8.15
2	2114	0.35	564	197.4
1.6	2002	-0.05	452	-22.6
1	937	-0.65	-613	398.45
1.4	869	-0.25	-681	170.25
			Total	1703.6

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{1703.6}{(12-1)(0.3317)(559.5009)} \\ &\approx 0.8346 \end{aligned}$$

Step 3

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.8346\frac{559.5009}{0.3317} = 1407.9339$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 1550 - 1407.9339 \cdot 1.65 = -773.0909$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -773.0909 + 1407.9339x$$

Step 4

(b) We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y - \hat{y})$ for each given y-value, which is the residual.

X	Y	$\hat{Y}$	$Y - \hat{Y}$
2	2017	2042.7769	-25.7769
1.6	922	1479.6033	-557.6033
1.6	1064	1479.6033	-415.6033
1.8	1942	1761.1901	180.8099
2	2137	2042.7769	94.2231
1.2	1012	916.4298	95.5702
2	2197	2042.7769	154.2231
1.6	1387	1479.6033	-92.6033
2	2114	2042.7769	71.2231
1.6	2002	1479.6033	522.3967
1	937	634.8430	302.1570
1.4	869	1198.0166	-329.0166

We note that -557.6033 is the most extreme residual (residual that is largest in absolute value), which implies that the second computer is much cheaper than expected and thus underpriced.

Step 5

(c) By part (a): $r = 0.8346$

Given claim: Correlation is greater than 0

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 12 - 2 = 10$ and in the column with $\alpha = 0.05$.

$$t = 1.812$$

The rejection region then contains all values greater than 1.812 and thus we reject H_0 when the t-value is larger than 1.812.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.8346\sqrt{12-2}}{\sqrt{1-0.8346^2}} \approx 4.7912$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 4.7912 > 1.812 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the population correlation is greater than zero.

Result

(a) $\hat{y} = -773.0909 + 1407.9339x$

(b) Yes, computer 2 is underpriced

(c) $r = 0.8346$, There is sufficient evidence to support the claim that the population correlation is greater than zero.

PROBLEM STATEMENT

Chapter 13, Page 506

Step 1

Given:

$$n = \text{Sample size} = 20$$
$$\alpha = \text{Significance level} = 0.05$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{1500 + 750 + 1500 + ... + 500}{20} = 1212.5$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{205 + 70 + 199 + ... + 44}{20} = 157.95$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(1500 - 1212.5)^2 + ... + (500 - 1212.5)^2}{20 - 1}} \approx 446.2932$$
$$s_y = \sqrt{\frac{(205 - 157.95)^2 + ... + (44 - 157.95)^2}{20 - 1}} \approx 70.7445$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X-\overline{X}$	$Y-\overline{Y}$	$(X-\overline{X})(Y-\overline{Y})$
1500	205	287.5	47.05	13526.875
750	70	-462.5	-87.95	40676.875
1500	19	287.5	41.05	11801.875
....	...	...	...	...
500	44	-712.5	-113.95	81189.375
			Total	563462.5

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{563462.5}{(20-1)(446.2932)(70.7445)} \\ &\approx 0.9393 \end{aligned}$$

Step 3

(b) Given claim: Correlation is greater than 0

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 4

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 20 - 2 = 18$ and in the column with $\alpha = 0.05$.

$$t = 1.734$$

The rejection region then contains all values greater than 1.734 and thus we reject H_0 when the t-value is larger than 1.734.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9393\sqrt{20-2}}{\sqrt{1-0.9393^2}} \approx 11.6151$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 11.6151 > 1.734 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the population correlation is greater than zero.

Step 5

(c) **Regression equation**

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9393\frac{70.7445}{446.2932} = 0.1489$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 157.95 - 0.1489 \cdot 1212.5 = -22.5813$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -22.5813 + 0.1489x$$

Step 6

(d) We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y - \hat{y})$ for each given y-value, which is the residual.

X	Y	$\hat{Y}$	$Y - \hat{Y}$
1500	205	20.7687	4.2313
750	70	89.0937	-19.0937
1500	19	200.7687	-12.5437
1250	151	163.5437	-12.5437
1250	181	163.5437	17.4563
1250	217	163.5437	53.4563
1000	94	126.3187	-32.3187
2000	298	275.2187	22.7813
1000	135	126.3187	8.6813
1500	211	200.7687	10.2313
1250	116	163.5437	-47.5437
500	72	51.8687	20.1313
500	82	51.8687	30.1313
1500	206	200.7687	5.2313
2000	245	275.2187	-30.2187
1500	219	200.7687	18.2313
750	63	89.0937	-26.0937
1500	200	200.7687	-0.7687
500	44	51.8687	-7.8687

The best-buy will have the highest residual, because this heater will then cover a much higher area than expected.

In this case, we note that the highest residual is 53.4563 and thus heater 6 is the best buy.

Result

(a) $r = 0.9393$

(b) Reasonable

(c) $\hat{y} = -22.5813 + 0.1489x$

(d) Heater 6

PROBLEM STATEMENT

Chapter 13, Page 507

Step 1

Given:

$$n = \text{Sample size} = 18$$
$$\alpha = \text{Significance level} = 0.05$$

(a) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{41 + 148 + 79 + ... + 168}{18} = 92.6111$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{3 + 8 + 5 + ... + 9}{18} = 5.3333$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(41 - 92.6111)^2 + ... + (168 - 92.6111)^2}{18 - 1}} \approx 47.1395$$
$$s_y = \sqrt{\frac{(3 - 5.3333)^2 + ... + (9 - 5.3333)^2}{18 - 1}} \approx 2.0292$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X-\overline{X}$	$Y-\overline{Y}$	$(X-\overline{X})(Y-\overline{Y})$
41	3	-51.6111	-2.3333	120.4242
148	8	55.3889	2.6667	147.7056
79	5	-13.6111	-0.3333	4.5366
....	...	...	...	...
168	9	75.3889	3.6667	276.4285
			Total	1605.3333

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{1605.3333}{(18-1)(47.1395)(2.0292)} \\ &\approx 0.9872 \end{aligned}$$

Step 3

Given claim: Correlation is greater than 0

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 4

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 18 - 2 = 16$ and in the column with $\alpha = 0.05$.

$$t = 1.746$$

The rejection region then contains all values greater than 1.746 and thus we reject H_0 when the t-value is larger than 1.746.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9872\sqrt{18-2}}{\sqrt{1-0.9872^2}} \approx 24.7594$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 24.7594 > 1.746 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the population correlation is greater than zero.

Step 5

(b) **Regression equation**

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9872\frac{2.0292}{47.1395} = 0.0425$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 5.3333 - 0.0425 \cdot 92.6111 = 1.3977$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 1.3977 + 0.0425x$$

Step 6

(c) We determine the estimated value $\hat{y}$ of the regression equation for each given value.

Next, we determine the value $(y - \hat{y})$ for each given y-value, which is the residual.

X	Y	$\hat{Y}$	$Y - \hat{Y}$
41	3	3.1402	-0.1402
148	8	7.6877	0.3123
79	5	4.7552	0.2448
41	4	3.1402	0.8598
85	5	5.0102	-0.0102
111	6	6.1152	-0.1152
37	3	2.9702	0.0298
111	6	6.1152	-0.1152
41	3	3.1402	-0.1402
91	5	5.2652	-0.2652
109	6	6.0302	-0.0302
207	10	10.1952	-0.1952
49	3	3.4802	-0.4802
113	6	6.2002	-0.2002
84	5	4.9677	0.0323
95	5	5.4352	-0.4352
57	4	3.8202	0.1798
168	9	8.5377	0.4623

The largest residual in absolute value is 0.8598, which belongs to dog 4 and thus dog 4 appears to be an overeater.

Result

(a) $r = 0.9872$, Reasonable

(b) $\hat{y} = 1.3977 + 0.0425x$

(c) Dog 4 is an overeater

PROBLEM STATEMENT

Chapter 13, Page 507

Step 1

(a) General equation of the least-squares regression line:

$$\hat{y} = a + bx$$

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the regression output:

$$a = 12.3601$$

The estimate of the slope b is given in the row "Distance-X" and in the column "Coefficients" of the regression output:

$$b = 4.7956$$

Replace a by 12.3601 and b by 4.7956 in the general equation of the least-squares regression line:

$$\hat{y} = a + bx = 12.3601 + 4.7956x$$

where x represents the distance from the fire station and y represents the amount of fire damage (in thousands of dollars).

Step 2

(b) Evaluate the regression line from part (a) at $x = 5$:

$$\hat{y} = a + bx = 12.3601 + 4.7956(5) = 36.3381$$

Thus we would estimate a damage of 36.3381 thousand dollars or $\$36,338.10$.

Step 3

(c) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 1,864.5782$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 3,209.0716$$

The coefficient of determination is the ratio of the regression sum of squares and the total sum of squares.

$$r^2 = \frac{SSR}{SS\ total} = \frac{1,864.5782}{3,209.0716} \approx 0.5810 = 58.10\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

58.10% of the variation in the amount of fire damage is explained by the variation in the distance from the fire station.

Step 4

(d) The estimate of the slope b is given in the row "Distance-X" and in the column "Coefficients" of the regression output:

$$b = 4.7956$$

Since the slope $b = 4.7956$ is positive, the correlation coefficient needs to be positive as well.

The correlation coefficient is then the positive square root of the coefficient of determination (which was determined in part (a)):

$$r = +\sqrt{r^2} = +\sqrt{0.5810} \approx 0.7622$$

Step 5

(e) Given:

$$\alpha = \text{Significance level} = 0.01$$

The estimate of the slope b is given in the row "Distance-X" and in the column "Coefficients" of the regression output:

$$b = 4.7956$$

The standard deviation of the slope b is given in the row "Distance-X" and in the column "Std. Error" of the regression output:

$$s_b = 0.7696$$

Given claim: Slope is nonzero (significant relationship)

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

Step 6

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value t_0 that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.005$$

The error degrees of freedom is given in the row "Residual" and in the column "df" of the given ANOVA table:

$$d.f._E = 28$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = d.f._E = 28$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 2.763$$

The rejection region then contains all values smaller than -2.763 and all values greater than 2.763, thus we reject H_0 when $t < -2.763$ or $t > 2.763$.

Compute the value of the test statistic:

$$t = \frac{b - 0}{s_b} = \frac{4.7956 - 0}{0.7696} \approx 6.2313$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 6.2313 > 2.763 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a significant relationship between the distance from the fire station and the amount of damage.

Result

(a) $\hat{y} = 12.3601 + 4.7956x$

(b) 36.3381 thousand dollars

(c) 58.10% of the variation in the amount of fire damage is explained by the variation in the distance from the fire station.

(d) $r = 0.7622$

(e) Yes

PROBLEM STATEMENT

Chapter 13, Page 508

Step 1

Given:

$$n = \text{Sample size} = 50$$

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2729.7 + 1835 + ... + 469.7}{50} = 806.59$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{237 + 489.3 + ... + 285.1}{50} = 386.1996$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(2729.7 - 806.59)^2 + ... + (469.7 - 806.59)^2}{50 - 1}} \approx 366.1071$$
$$s_y = \sqrt{\frac{(237 - 386.1996)^2 + ... + (285.1 - 386.1996)^2}{50 - 1}} \approx 173.2606$$

Step 2

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X-\overline{X}$	$Y-\overline{Y}$	$(X-\overline{X})(Y-\overline{Y})$
2729.7	237	1923.11	-149.1996	-286927.2428
1835	789.3	1028.41	403.1004	414552.4824
1129.2	377	322.61	-9.1996	-2967.8830
...	...	....	...	...
469.7	285.1	-336.89	-101.0996	34059.4442
			Total	86725.6218

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X-\overline{X})(Y-\overline{Y})}{(n-1)s_x s_y} \\ &= \frac{86725.6218}{(50-1)(366.1071)(173.2606)} \\ &\approx 0.0279 \end{aligned}$$

We note that the correlation coefficient is nearly 0, which indicates that there appears to be no linear association between the two variables.

Moreover, this then also implies that movies with large budgets do not necessarily result in large box office revenues.

Result

$r = 0.0279$, No

PROBLEM STATEMENT

Chapter 13, Page 509

Step 1

(a) I will use Excel to determine the linear regression model between size of home and selling price. The corresponding output is given in the image below:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.3710415948
R Square	0.1376718651
Adjusted R Squa	0.1292997472
Standard Error	43.95467488
Observations	105

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	31770.20439	31770.20439	16.44409074	0.000097560657
Residual	103	198997.3848	1932.013444		
Total	104	230767.5891			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	64.79312352	38.78409957	1.670610488	0.09783470026	-12.12598842	141.7122355
X Variable 1	0.07028917358	0.01733338984	4.055131409	0.000097560658	0.03591248394	0.1046658632

Step 2

Regression equation

The estimate of the slope b is given in the row "X Variable 1" and in the column "Coefficients" of the output:

$$b \approx 0.0703$$

The estimate of the y-intercept is given in the row "Intercept" and in the column "Coefficients" of the output:

$$a \approx 64.7931$$

The general linear regression equation is of the form $\hat{y} = a + bx$.

$$\hat{y} = a + bx = 64.7931 + 0.0703x$$

Step 3

95% confidence interval

Evaluate the regression equation at $x = 2200$:

$$\hat{y} = a + bx = 64.7931 + 0.0703(2200) = 219.4531$$

The standard error of the estimate is given to the right of "Standard Error" in the output:

$$s = 43.9547$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 105 - 2 = 103$ and $c = 95\%$:

$$t_c = \pm 1.983$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{\frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 1.983(43.9547)\sqrt{\frac{1}{105} + \frac{(2200 - 2223.8095)^2}{6430476.19}} \\ &\approx 8.5454 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 219.4531 - 8.5454 = 210.9077 \\ \hat{y} + E &= 219.4531 + 8.5454 = 227.9985 \end{aligned}$$

We are 95% confident that the mean selling price of homes with 2200 square feet is between 210.9077 thousand dollars and 227.9985 thousand dollars.

Step 4

95% prediction interval

Evaluate the regression equation at $x = 2200$:

$$\hat{y} = a + bx = 64.7931 + 0.0703(2200) = 219.4531$$

The standard error of the estimate is given to the right of "Standard Error" in the output:

$$s = 43.9547$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 105 - 2 = 103$ and $c = 95\%$:

$$t_c = \pm 1.983$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 1.983(43.9547)\sqrt{1 + \frac{1}{105} + \frac{(2200 - 2223.8095)^2}{6430476.19}} \\ &\approx 87.5801 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 219.4531 - 87.5801 = 131.8730 \\ \hat{y} + E &= 219.4531 + 87.5801 = 307.0332 \end{aligned}$$

We are 95% confident that the selling price of a home with 2200 square feet is between 131.8730 thousand dollars and 307.0332 thousand dollars.

Step 5

(b) I will use Excel to determine the linear regression model between distance from center of the city and selling price.

The corresponding output is given in the image below:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.3470311664
R Square	0.1204306305
Adjusted R Squa	0.111891122
Standard Error	44.39191115
Observations	105

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	27791.48625	27791.48625	14.10275911	0.000286916225
Residual	103	202976.1029	1970.641776		
Total	104	230767.5891			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	270.1669889	13.76460436	19.62766106	0	242.8681434	297.4658345
X Variable 1	-3.353993385	0.8931207054	-3.755364045	0.000286916225	-5.125287608	-1.582699162

Step 6

Regression equation

The estimate of the slope b is given in the row "X Variable 1" and in the column "Coefficients" of the output:

$$b \approx -3.3540$$

The estimate of the y-intercept is given in the row "Intercept" and in the column "Coefficients" of the output:

$$a \approx 270.1670$$

The general linear regression equation is of the form $\hat{y} = a + bx$.

$$\hat{y} = a + bx = 270.1670 - 3.3540x$$

Step 7

95% confidence interval

Evaluate the regression equation at $x = 20$:

$$\hat{y} = a + bx = 270.1670 - 3.3540(20) = 203.087$$

The standard error of the estimate is given to the right of "Standard Error" in the output:

$$s = 44.3919$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 105 - 2 = 103$ and $c = 95\%$:

$$t_c = \pm 1.983$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{\frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 1.983(44.3919)\sqrt{\frac{1}{105} + \frac{(20 - 14.6286)^2}{518880913.4}} \\ &\approx 8.5908 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 203.087 - 8.5908 = 194.4962 \\ \hat{y} + E &= 203.087 + 8.5908 = 211.6778 \end{aligned}$$

We are 95% confident that the mean selling price of homes 20 miles from the center of the city is between 194.4962 thousand dollars and 211.6778 thousand dollars.

Step 8

95% prediction interval

valuate the regression equation at $x = 20$:

$$\hat{y} = a + bx = 270.1670 - 3.3540(20) = 203.087$$

The standard error of the estimate is given to the right of "Standard Error" in the output:

$$s = 44.3919$$

Determine the critical value using the Student's T distribution table in the appendix with $df = n - 2 = 105 - 2 = 103$ and $c = 95\%$:

$$t_c = \pm 1.983$$

Determine the margin of error:

$$\begin{aligned} E &= t_c s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{x})^2}{\sum (x - \overline{x})^2}} \\ &= 1.983(44.3919)\sqrt{1 + \frac{1}{105} + \frac{(20 - 14.6286)^2}{518880913.4}} \\ &\approx 88.4473 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\begin{aligned} \hat{y} - E &= 203.087 - 88.4473 = 114.6397 \\ \hat{y} + E &= 203.087 + 88.4473 = 291.5343 \end{aligned}$$

We are 95% confident that the selling price of a home 20 miles from the center of the city is between 114.6397 thousand dollars and 291.5343 thousand dollars.

Step 9

(c) **Hypothesis test correlation Distance from city**

Given claim: Negative correlation

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \geq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho < 0 \text{ (There is a negative linear correlation)}$$

The coefficient of determination R^2 is mentioned to the right of "R Square" in the output:

$$R^2 = 0.1204$$

Since the slope b is negative, the correlation coefficient r is the negative square root of the coefficient of determination:

$$r = -\sqrt{R^2} = -\sqrt{0.1204} \approx -0.3470$$

Step 10

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left -tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value t_0 that has a probability of 0.05 to its left.

$$P(t < t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 105 - 2 = 103$ and in the column with $\alpha = 0.05$.

$$t = -1.660$$

The rejection region then contains all values smaller than -1.660 and thus we reject H_0 when the t-value is smaller than -1.660.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.3470\sqrt{105-2}}{\sqrt{1-(-0.3470)^2}} \approx -3.755$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = -3.755 < -1.660 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a negative correlation between distance from the center of the city and selling price. By the output, the corresponding P-value is 0.0002869 (given in the row X Variable 1 and in the column P-value of the output).

Step 11

Hypothesis test correlation Area home

Given claim: Positive correlation

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \geq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

The coefficient of determination R^2 is mentioned to the right of "R Square" in the output:

$$R^2 = 0.1293$$

Since the slope b is positive, the correlation coefficient r is the positive square root of the coefficient of determination:

$$r = \sqrt{R^2} = \sqrt{0.1293} \approx 0.3596$$

Step 12

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value below the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 105 - 2 = 103$ and in the column with $\alpha = 0.05$.

$$t = 1.660$$

The rejection region then contains all values larger than 1.660 and thus we reject H_0 when the t-value is larger than 1.660.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.3596\sqrt{105-2}}{\sqrt{1-0.3596^2}} \approx 3.911$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 3.911 > 1.660 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a positive correlation between area of the home and selling price. By the output, the corresponding P-value is 0.00009756 (given in the row X Variable 1 and in the column P-value of the output).

Result

(a) $\hat{y} = 64.7931 + 0.0703x$, 219.4531 thousand dollars, $(210.9077, 227.9985)$, $(131.8730, 307.0332)$

(b) $\hat{y} = 270.1670 - 3.3540x$, 203.087 thousand dollars, $(194.4962, 211.6778)$, $(114.6397, 291.5343)$

(c) Yes

PROBLEM STATEMENT

Chapter 13, Page 509

Step 1

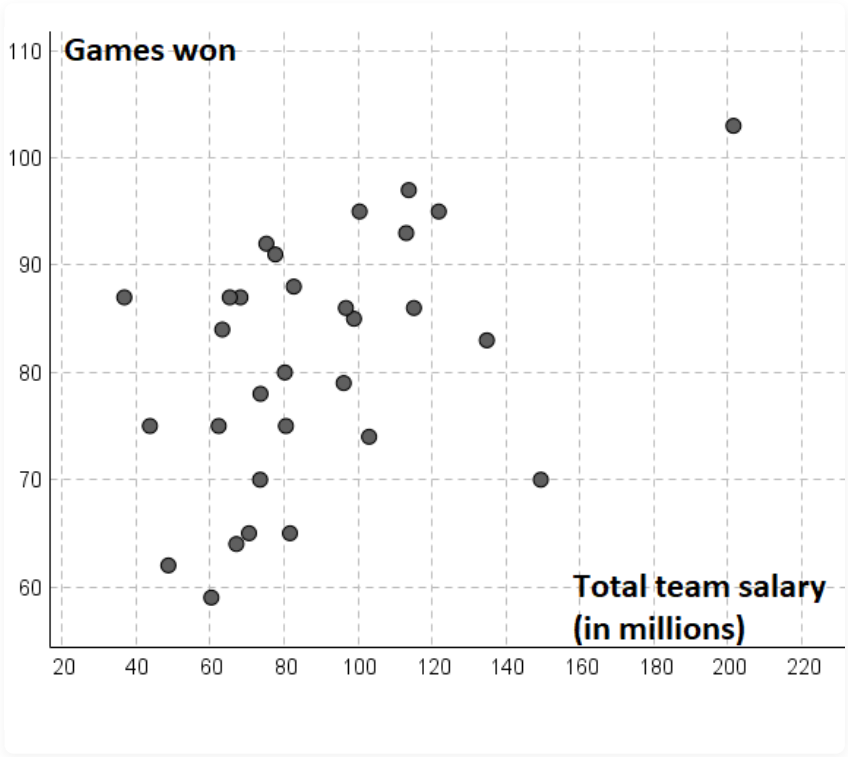
Given:

$$n = \text{Sample size} = 30$$

(a)

Scatter diagram

Total team salary is on the horizontal axis and the number of games won is on the vertical axis.



Step 2

The pattern in the scatterplot slopes upwards, which indicates that there is a direct relationship between the two variables.

Step 3

(b)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{62.3 + 36.8 + 48.7 + ... + 100.4}{30} = 88.5133$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{75 + 87 + 62 + ... + 95}{30} = 81$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(62.3 - 88.5133)^2 + ... + (100.4 - 88.5133)^2}{30 - 1}} \approx 33.9011$$
$$s_y = \sqrt{\frac{(75 - 81)^2 + ... + (95 - 81)^2}{30 - 1}} \approx 11.4350$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
62.3	75	-26.2133	-6	157.2798
36.8	87	-51.7133	6	-310.2798
48.7	62	-39.8133	-19	756.4527
...	...	...	...	...
100.4	95	11.8867	14	166.4138
			Total	**5225**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{5225}{(30-1)(33.9011)(11.4350)} \\ &\approx 0.4648 \end{aligned}$$

Step 5

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.4648\frac{11.4350}{33.9011} = 0.1568$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 81 - 0.1568 \cdot 88.5133 = 67.1238$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 67.1238 + 0.1568x$$

Evaluate the regression line at $x = 100.0$:

$$\hat{y} = a + bx = 67.1238 + 0.1568(100.0) = 82.8008$$

Thus we would estimate 82.8008 wins with a salary of $\$100.0$ million.

Step 6

(c) Result part (b): $b = 0.1568$

The estimate increase in the number of wins when the salary increases by $\$5$ million is the product of 5 and the slope:

$$5b = 5(0.1568) = 0.784$$

Thus we would estimate 0.784 additional wins when there is an additional $\$5$ million in the salary.

Step 7

(d) Since we have not been given a formula for the calculation of the standard deviation of the slope, I will use the hypothesis test for the correlation coefficient instead (which will have the same results as a hypothesis test for the slope).

Given claim: Slope is positive

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 8

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 30 - 2 = 28$ and in the column with $\alpha = 0.05$.

$$t = 1.701$$

The rejection region then contains all values greater than 1.701 and thus we reject H_0 when the t-value is larger than 1.701.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.4648\sqrt{30-2}}{\sqrt{1-0.4648^2}} \approx 2.778$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 2.778 > 1.701 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the slope of the regression line is positive.

Step 9

(e)

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = (0.4648)^2 \approx 0.2160 = 21.60\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

21.60% of the variation in the number of wins is explained by the variation in the team salary.

Step 10

(f)

Wins and team batting average

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{62.3 + 36.8 + 48.7 + ... + 100.4}{30} = 88.5133$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{0.262 + 0.268 + 0.252 + ... + 0.27}{30} = 0.2623$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(62.3 - 88.5133)^2 + ... + (100.4 - 88.5133)^2}{30 - 1}} \approx 33.9011$$
$$s_y = \sqrt{\frac{(0.262 - 0.2623)^2 + ... + (0.27 - 0.2623)^2}{30 - 1}} \approx 0.0090$$

Step 11

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
62.3	0.262	-26.2133	-0.0003	0.0079
36.8	0.268	-51.7133	0.0057	-0.2948
48.7	0.252	-39.8133	-0.0103	0.4101
...	...	...	...	...
100.4	0.27	11.8867	0.0077	0.0915
			Total	**4.3287**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{4.3287}{(30-1)(33.9011)(0.0090)} \\ &\approx 0.4876 \end{aligned}$$

Step 12

Wins and team ERA

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{62.3 + 36.8 + 48.7 + ... + 100.4}{30} = 88.5133$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{4.26 + 4.26 + 4.59 + ... + 3.41}{30} = 4.314$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(62.3 - 88.5133)^2 + ... + (100.4 - 88.5133)^2}{30 - 1}} \approx 33.9011$$
$$s_y = \sqrt{\frac{(4.26 - 4.314)^2 + ... + (3.41 - 4.314)^2}{30 - 1}} \approx 0.4313$$

Step 13

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
62.3	4.26	-26.2133	-0.054	1.4155
36.8	4.29	-51.7133	-0.024	1.2411
48.7	4.59	-39.8133	0.276	-10.9885
....	...	...	...	...
100.4	3.41	11.8867	-0.904	-10.7456
			Total	**-99.8776**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{-99.8776}{(30-1)(33.9011)(0.4313)} \\ &\approx -0.2356 \end{aligned}$$

We note that the correlation coefficient $r = 0.4876$ is further from zero than the correlation coefficient $r = -0.2356$, which implies that the correlation between wins and team batting average is stronger.

Result

(a) Yes

(b) 82.8008 wins

(c) 0.784 additional wins

(d) Yes

(e) 21.60%

(f) Wins and team batting average

PROBLEM STATEMENT

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Step 1

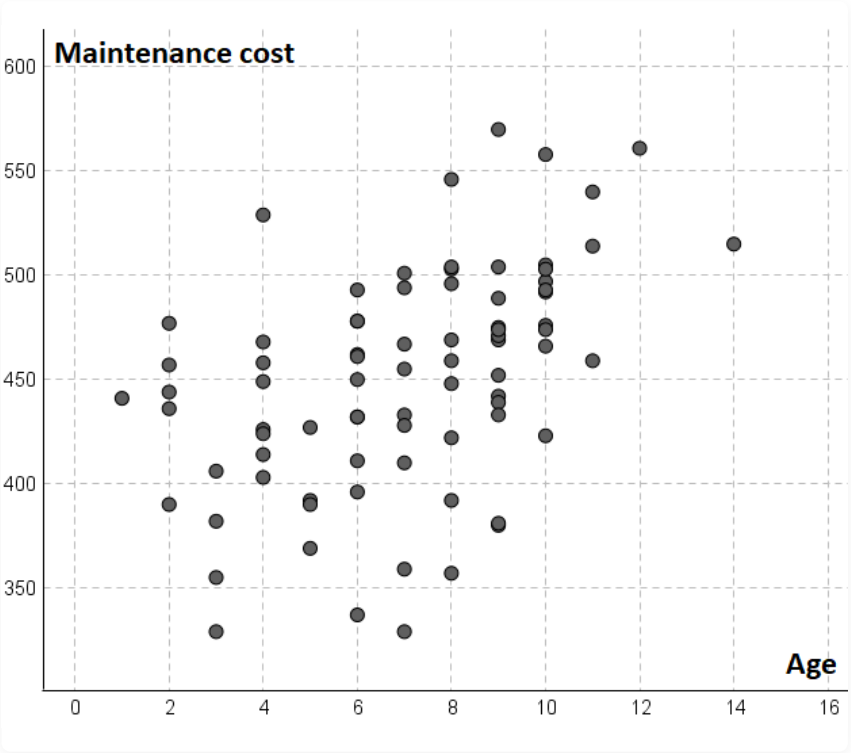
Given:

$$n = \text{Sample size} = 80$$

(a)

Scatter diagram

Age is on the horizontal axis and the maintenance cost is on the vertical axis.



Step 2

The pattern in the scatterplot slopes upwards, which indicates that there is a direct relationship between the two variables.

The relation appears to be weak, because the points appear to be spread very far apart in the pattern.

Step 3

(b)

Correlation coefficient

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{6 + 6 + 4 + ... + 10}{80} = 7$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{337 + 396 + 403 + ... + 493}{80} = 450.2875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(6-7)^2 + ... + (10-7)^2}{80-1}} \approx 2.7513$$
$$s_y = \sqrt{\frac{(337-450.2875)^2 + ... + (493-450.2875)^2}{80-1}} \approx 53.6860$$

Step 4

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
6	337	-1	-113.2875	113.2875
6	396	-1	-54.2875	54.2875
4	403	-3	-47.2875	141.8625
...	...	...	...	...
10	493	3	42.7125	128.1375
			Total	**5424**

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum (X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{5424}{(80-1)(2.7513)(53.6860)} \\ &\approx 0.4648 \end{aligned}$$

Step 5

Regression equation

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.4648\frac{53.6860}{2.7513} = 9.0702$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 450.2875 - 9.0702 \cdot 7 = 386.7959$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = 386.7959 + 9.0702x$$

Evaluate the regression line at $x = 10$:

$$\hat{y} = a + bx = 386.7959 + 9.0702(10) = 477.4982$$

Thus we would estimate a maintenance cost of about $\$477.50$ for a 10-year old bus.

Step 6

(c) Since we have not been given a formula for the calculation of the standard deviation of the slope, I will use the hypothesis test for the correlation coefficient instead (which will have the same results as a hypothesis test for the slope).

Given claim: Slope is greater than zero

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \leq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho > 0 \text{ (There is a positive linear correlation)}$$

Step 7

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 80 - 2 = 78$ and in the column with $\alpha = 0.05$.

$$t = 1.665$$

The rejection region then contains all values greater than 1.665 and thus we reject H_0 when the t-value is larger than 1.665.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.4648\sqrt{80-2}}{\sqrt{1-0.4648^2}} \approx 4.636$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 4.636 > 1.665 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the slope of the regression line is greater than zero.

Result

(a) Direct, weak

(b) $\$477.50$

(c) There is sufficient evidence to support the claim that the slope of the regression line is greater than zero.